

האוניברסיטה העברית בירושלים
הפקולטה למתמטיקה ולמדעי הטבע
המכון למתמטיקה

עבודת גמר לתואר מוסמך בנושא:

התפלגות נקודות רציונליות על עקומים אליפטיים

**Distribution of I -Sylow subgroups of
rational points of elliptic curves over
fields of prime order**

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העבודה הוכנה בהדרכת פרופ' ארי שנידמן

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תקציר: התפלגות נקודות רציונליות על עקומים אליפטיים

גיא ספיר

יהי E עקום אליפטי מעל $\mathbb{Q}$. מהי צפיפות קבוצת המספרים הראשוניים עם "צמצום טוב" עבורם $E(\mathbb{F}_p) \cong G$ לאיזושהי חבורה אבלית נתונה G ? מהחסם של הסה התשובה תמיד תהיה 0, ולכן השאלה חסרת תוכן.

במקום זאת, נקבע איזושהו ראשוני $\ell \in \mathbb{N}$. לכל ראשוני p כך של- E יש "צמצום טוב", $E(\mathbb{F}_p)[\ell^\infty]$ החלק ה- ℓ של חבורות הנקודות ה- F_p -רציונליות של E היא חבורת- ℓ מדרגה לכל היותר 2.

בתזה זאת אנחנו נחקור את ההסתברות שחבורה קבועה G מופיעה בתור $E(\mathbb{F}_p)[\ell^\infty]$ כאשר עוברים על ראשוניים p עם "צמצום טוב".

נחיל עם התורה הכללית, הסובבת סביב תפקידיהם של העתקות "פרובניוס" השונות. אנו מדמים את החבורה $E(\mathbb{F}_p)[\ell^\infty]$ כאשר p משתנה על ידי קורעין של מטריצה אקראית ב- $M_2(\mathbb{Z}_\ell)$ תחת איזושהי התפלגות טבעית. בפרט, אנו מוכיחים את המשפט המרכזי של חלק זה, שהוא המשוואה

$$\mathbb{P}(p \in C, E(\mathbb{F}_p)[\ell^\infty] \cong G) = [GL_2(\mathbb{Z}_\ell) : (\rho_\ell)] \cdot \mu \left(\left\{ M \in (\rho_\ell) \mid \det(M) \in C, \mathbb{Z}_\ell^2 / (1 - M)\mathbb{Z}_\ell^2 \cong G \right\} \right)$$

כאשר E עקום אליפטי ללא "כפל מרוכב", G חבורת- ℓ , $C \subseteq \mathbb{N}$, μ מידת האר הסתברות על $GL_2(\mathbb{Z}_\ell)$ ו- $\rho_\ell : G_{\mathbb{Q}} \rightarrow GL_2(\mathbb{Z}_\ell)$ ההצגה ה- ℓ -אדית המוצמדת ל- E . משמעות אגף שמאל במשוואה היא צפיפות הראשוניים המקיימים את התנאי בסגוריים.

לאחר מכן, נתעמק בחישובים ונבסס אסטרטגיה לחישוב הסתברויות מותנות מהצורה

$$\mathbb{P}(E(\mathbb{F}_p)[\ell^\infty] \cong G \mid v_\ell(p-1) = t)$$

עבור t קבוע, על ידי צמצום הבעיה לבעית מניה קומבינטורית. נשתמש באסטרטגיה זאת בשביל לחשב כמה מקרים פרטיים.

לבסוף, נפרש את התוצאות במסגרת היוריסטיקות כהן-לנסטרה ובכך נשער השערה כללית המבוססת על החישובים שמופיעים בחלק הקודם. בנוסף, נזכיר גם עבודות שונות העלולות לבסס דרכים שונות להגיע לאותן התוצאות, בפרט שיטת המומנטים שנופצה במספר עבודות לאחרונה ובוססה במסגרת פורמלית על ידי סווינזוד.

Distribution of ℓ -Sylow subgroups of rational points of elliptic curves over fields of prime order

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Abstract

Given an elliptic curve E defined over $\mathbb{Q}$ without CM and an abelian group G , what is the density of primes of good reduction p such that $E(F_p) \cong G$? From Hasse's bound this will come out to be 0 always, so this is not a good question. Instead we fix a prime ℓ and ask for the density of primes p such that $E(\mathbb{F}_p)[\ell^\infty] \cong G$. We solve this question in several specific cases and establish a strategy to calculate the conditional probability restricting to primes such that $v_\ell(p-1)$ is fixed. In order to do so, we model the group $E(\mathbb{F}_p)[\ell^\infty]$ as p varies by a cokernel of a random matrix in $GL_2(\mathbb{Z}_\ell)$

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1 Introduction

Let $\ell \in \mathbb{N}$ be a prime and E an elliptic curve over $\mathbb{Q}$. If E has good reduction at a prime p , then $E(\mathbb{F}_p)[\ell^\infty]$, the ℓ -part of the $\mathbb{F}_p$ -rational points of E , is an abelian ℓ -group of rank ≤ 2 . In this paper we investigate the probability that a fixed group G appears as $E(\mathbb{F}_p)[\ell^\infty]$. Since E has good reduction at all but finitely many primes, we can ask what is the density of primes p such that $E(\mathbb{F}_p)[\ell^\infty] \cong G$, disregarding bad primes.

In Section 3 we develop the general theory. For a set X of prime numbers the notation $\mathbb{P}(X)$ stands for the natural density of X in the set of all primes. For example, in the theorem below the left hand side stands for the density of primes p such that $p \in C$ and $E(\mathbb{F}_p)[\ell^\infty] \cong G$. In particular, we prove the main theorem of this section, which is

Theorem. *Assume E does not have CM. Then for every abelian ℓ -group G and every subset $C \subseteq \mathbb{N}$:*

$$\mathbb{P}(p \in C, E(\mathbb{F}_p)[\ell^\infty] \cong G) = [GL_2(\mathbb{Z}_\ell) : \text{im}(\rho_\ell)] \cdot \mu \left(\left\{ M \in \text{im}(\rho_\ell) \mid \det(M) \in C, \frac{\mathbb{Z}_\ell^2}{(1-M)\mathbb{Z}_\ell^2} \cong G \right\} \right)$$

Where μ is the probability Haar measure on $GL_2(\mathbb{Z}_\ell)$ and $\rho_\ell : G_\mathbb{Q} \rightarrow GL_2(\mathbb{Z}_\ell)$ is the ℓ -adic representation attached to E .

The assumption that E does not have CM is needed for Serre's open image theorem (Theorem 2), which also states that for all but finitely many ℓ ρ_ℓ is surjective and so the right hand side essentially does not depend on E . So the theorem can be interpreted as saying that $E(\mathbb{F}_p)[\ell^\infty]$ is "distributed" by this natural distribution for abelian ℓ -groups of rank ≤ 2 .

In Section 4 we delve into computations and develop a plan to compute conditional probabilities of the form $\mathbb{P}(E(\mathbb{F}_p)[\ell^\infty] \cong G \mid v_\ell(p-1) = t)$ for a fixed t , essentially reducing the problem to a combinatorial counting problem. We use this plan to compute several specific cases.

In Section 5 we recap the results, interpret them in the context of the Cohen-Lenstra heuristics, thus conjecturing a general statement and finally mention other works which might establish different methods to come by results similar to ours, namely the moment method as formalized in [SW22].

2 Preliminaries

2.1 Elliptic Curves and their ℓ -adic Representations

This is standard material, as appears in [Sil09] and [Ser68], for example.

Definition 1 (Elliptic Curve). An elliptic curve E over a field $\mathbb{F}$ is a smooth, projective, geometrically connected algebraic curve over $\mathbb{F}$ of genus 1 with a choice of a distinguished point O . Using Riemann-Roch, one can show that every elliptic curve over a field of characteristic different from 2, 3 has an affine model of the form:

$$E : y^2 = x^3 + ax + b$$

for some $a, b \in \mathbb{F}$. Here O is the point at infinity.

It is well known that E is an abelian variety, meaning it has the structure of an abelian group such that O is the identity element and the multiplication and inversion maps are morphisms of algebraic varieties.

As an abelian group, for every $n \in \mathbb{Z}$ E has a multiplication by n , denote by $[n] : E \rightarrow E$, defined by iterated addition (or subtraction if n is negative) and so $\mathbb{Z} \subseteq \text{End}(E)$.

Definition 2. If $Z = \text{End}(E)$, we say that E does not have complex multiplication (or CM), if $Z \subsetneq \text{End}(E)$ we say that E has complex multiplication.

Definition 3 (ℓ -adic Tate module of elliptic curve). Given an elliptic curve E over $\mathbb{F}$, we define an inverse system with maps $E(\overline{\mathbb{F}})[\ell^{n+1}] \xrightarrow{[\ell]} E(\overline{\mathbb{F}})[\ell^n]$. Define the ℓ -adic Tate module of E , denoted by $T_\ell E$, to be the inverse limit. Note that $E(\overline{\mathbb{F}})[\ell^n] \cong \mathbb{Z}/\ell^n \mathbb{Z} \times \mathbb{Z}/\ell^n \mathbb{Z}$ and so $T_\ell E \cong \mathbb{Z}_\ell^2$.

Definition 4 (The ℓ -adic representation attached to an elliptic curve). The action of the absolute Galois group $G_{\mathbb{F}}$ on points of $E(\overline{\mathbb{F}})$ coordinate-wise sends ℓ^n torsion points to ℓ^n torsion points. From this we get a continuous group action $G_{\mathbb{F}} \curvearrowright T_\ell E$ or in other words a representation $\rho_\ell : G_{\mathbb{F}} \rightarrow \text{Aut}(T_\ell E) \cong GL_2(\mathbb{Z}_\ell)$ which we call the ℓ -adic representation attached to E .

Definition 5. Given an elliptic curve E over $\mathbb{Q}$ and a prime p , we say that $T_\ell E$ is unramified at p if the image of the inertia subgroup $I_p \subseteq G_{\mathbb{Q}}$ under ρ_ℓ is trivial, i.e: $\rho_\ell(I_p) = 0$.

In this paper we are dealing with distribution of groups when we fix an elliptic curve and "vary" the field over which the curve is defined.

Definition 6 (Reduction of $E \bmod p$). (For more general and precise treatment see [Sil09] chapter 7) Given an elliptic curve E over $\mathbb{Q}$ and p a prime number, after choosing an appropriate model of E , we can reduce the equation modulo p and get a curve which we denote by E_p . This curve is

not necessarily an elliptic curve over $\mathbb{F}_p$, as it might be singular, but when it does happen to be an elliptic curve we say that p is a prime of good reduction.

Remark. When the meaning is obvious (in later sections of the paper), we will implicitly denote the $\mathbb{F}_p$ -rational points of E_p by $E(\mathbb{F}_p)$. This is fine in some sense because if E is an elliptic curve over $\mathbb{Q}$, the only way to interpret $E(\mathbb{F}_p)$ is as $E_p(\mathbb{F}_p)$.

It is well known, for example, that an elliptic curve over $\mathbb{Q}$ has good reduction for all but finitely many primes and that the primes of bad reduction (i.e not of good reduction) for a given elliptic curve can be found using the numerical invariant called the conductor of E .

The notion of good reduction relates to the ℓ -adic representation by the following criterion.

Theorem 1 (Neron-Ogg-Shafarevich Criterion, [Sil09] theorem 7.1). *Let E be an elliptic curve over $\mathbb{Q}$ and p be a prime number. Then E has good reduction at p if and only if $T_\ell E$ is unramified at p for ℓ a prime different from p ,*

Definition 7 (Frobenius Galois automorphism). Let $K/\mathbb{Q}$ be a Galois extension and $p \in \mathbb{Z}$ an unramified prime. For every prime $\mathfrak{p}$ over p the decomposition group $D_{\mathfrak{p}}$ is canonically isomorphic to $\text{Gal}(\mathbb{F}_{\mathfrak{p}}/\mathbb{F}_p)$ where $\mathbb{F}_{\mathfrak{p}}$ denotes the residue field $\mathcal{O}_K/\mathfrak{p}$ and so we get an element $Frob_p \in D_{\mathfrak{p}} \subseteq \text{Gal}(K/\mathbb{Q})$. Different choices of primes over p will define different $Frob_p$ conjugate to each other and so $Frob_p$ is uniquely defined up to conjugation.

Definition 8 (Frobenius isogeny). Let E be an elliptic curve over $\mathbb{Q}$ with good reduction at a prime p and denote by E_p the corresponding reduction, which is an elliptic curve over $\mathbb{F}_p$. The Frobenius isogeny $F_p : E_p \rightarrow E_p$ is defined by $(x, y) \mapsto (x^p, y^p)$.

We defined two maps with a very similar name, and so it is reasonable to find that they are compatible in the following sense.

Claim 1 (Compatibility of Frobenius maps). *Given an elliptic curve E over $\mathbb{Q}$ and a prime number ℓ , for every prime of good reduction p different from ℓ , the reduction map $E \rightarrow E_p$ induces an isomorphism $T_\ell E \xrightarrow{\sim} T_\ell E_p$ under which the action of $Frob_p$ on $T_\ell E$ via the ℓ -adic representation corresponds to the action on $T_\ell E_p$ induced by the isogeny F_p . In other words, the following diagram is commutative:*

$$\begin{array}{ccc} T_\ell E & \xrightarrow{Frob_p} & T_\ell E \\ \downarrow & & \downarrow \\ T_\ell E_p & \xrightarrow{F_p} & T_\ell E_p \end{array}$$

Proof. To prove commutativity of this diagram, it is enough to prove commutativity of the following

diagram

$$\begin{array}{ccc} E(\overline{\mathbb{Q}})[\ell^n] & \xrightarrow{Frob_p} & E(\overline{\mathbb{Q}})[\ell^n] \\ \downarrow & & \downarrow \\ E_p(\overline{\mathbb{F}_p})[\ell^n] & \xrightarrow{F_p} & E_p(\overline{\mathbb{F}_p})[\ell^n] \end{array}$$

for every n , then we can take inverse limit and get what we want. Commutativity of these diagram can be shown directly: If we take $(x, y) \in E(\overline{\mathbb{Q}})[\ell^n]$ then in one direction we get $F_p(\overline{x}, \overline{y})$ which is $(\overline{x^p}, \overline{y^p})$ (where the bars denote the reduction). In the other direction, we get $\overline{Frob_p(x, y)}$ and by definition $Frob_p$ is an automorphism that when reduced acts by raising to the p -th power, so $\overline{Frob_p(x, y)} = (\overline{x^p}, \overline{y^p})$ and we finished. $\square$

Remark. This compatibility is useful because on one side we have $Frob_p$ with its connections to algebraic number theory (for example Chebotarev's density theorem) and on the other hand we have F_p which is an isogeny with connections to geometry of E over $\mathbb{F}_p$.

Theorem 2 (Serre's Open Image Theorem, [Sil09] theorem 7.9). *If E is an elliptic curve over $\mathbb{Q}$ without CM, then for every prime ℓ , ρ_ℓ 's image is of finite index and furthermore for all but finitely many primes ℓ it is surjective.*

Theorem 3 (Chebotarev's Density Theorem for Infinite Extensions). *Let $K/\mathbb{Q}$ be some Galois extension (maybe infinite) unramified for all but finitely many primes and denote by μ the probability Haar measure on $\text{Gal}(K/\mathbb{Q})$. For every $X \subseteq \text{Gal}(K/\mathbb{Q})$ which is stable under conjugation, the density of primes p such that $Frob_p \in X$ equals $\mu(X)$.*

Proof. See [Ser68] chapter I, section 2.2, corollary 2. $\square$

Remark. Note that $Frob_p$ is only well-defined up to conjugation and so the statement of Theorem 3 makes sense only if X is stable under conjugation.

2.2 Calculating Haar measure via finite index subgroups

Let G be a compact second countable group, μ the probability Haar measure on G and $H \leq G$ some finite index subgroup. Any subset $X \subseteq G$ that can be written as a disjoint union of cosets $X = \coprod_{i \in I} a_i H$ will have measure:

$$\mu(X) = \sum_{i \in I} \mu(a_i H) = \sum_{i \in I} \underbrace{\mu(H)}_{= \frac{1}{[G:H]}} = \frac{\#I}{[G:H]}$$

We can reinterpret the fact that X is a disjoint union of cosets by saying that X is defined by some property "detected modulo H ", in the sense that if an element of g has that property (i.e g is in X) then all of g 's coset gH has that property (i.e gH is a subset of X).

Example 1. Denote by μ the probability Haar measure on $\mathbb{Z}_\ell$ and define

$$X := \{a \in \mathbb{Z}_\ell \mid v_\ell(a - 1) = 1\}$$

What is $\mu(X)$? We can answer with some symmetric reasoning or more formally writing down some transformations, but using the method explained above we can simply say that X is detected modulo ℓ^2 and X 's projection to $\mathbb{Z}_\ell/\ell^2\mathbb{Z}_\ell$ is $\{1 + k\ell\}_{k=1}^{\ell-1}$ and so $\mu(X) = \frac{\ell-1}{\ell^2}$.

Example 2. Denote by μ the probability Haar measure on $GL_2(\mathbb{Z}_5)$ and define

$$X := \left\{ M \in GL_2(\mathbb{Z}_5) \mid \begin{array}{l} \det(1 - M) \equiv 0 \pmod{5}, \\ \det((1 - M)^2) \equiv 0 \pmod{5} \end{array} \right\}$$

What is $\mu(X)$? Quite clearly X is detected modulo 5 (which is the same thing as projecting onto $GL_2(\mathbb{Z}_5)/1 + 5 \cdot M_2(\mathbb{Z}_5)$), as the conditions defining X are equivalences mod 5. From linear algebra, the projection of X to $GL_2(\mathbb{F}_5)$ is the set of matrices in $GL_2(\mathbb{F}_5)$ similar to the identity matrix I_2 , so $\overline{X} = \{I_2\}$ as this is the only such matrix and

$$\mu(X) = \frac{1}{\#GL_2(\mathbb{F}_5)} = \frac{1}{(5^2 - 1)(5^2 - 5)} = \frac{1}{480}$$

3 Conceptual Results

Claim 2. *Let E be an elliptic curve over $\mathbb{Q}$, p a prime of good reduction and $\ell \neq p$ a prime number.*

$$E(\mathbb{F}_p)[\ell^\infty] \cong T_\ell E / (1 - \text{Frob}_p) T_\ell E$$

Lemma 1. *The map $1 - F_p : T_\ell E_p \rightarrow T_\ell E_p$ is injective.*

Proof. $E_p(\mathbb{F}_p)$ is finite so there exists N such that $E_p(\mathbb{F}_p)[\ell^\infty] = E_p(\mathbb{F}_p)[\ell^N]$. $\ker(1 - F_p)$ consists of the elements of $T_\ell E_p$ whose coordinates are elements of $E_p(\mathbb{F}_p)[\ell^N]$, so if $(P_i)_{i=1}^\infty$ is in the kernel then for every i we have $P_i = [\ell^N]P_{N+i} = 0$. $\square$

We give an elementary proof of Claim 2, followed by a much nicer one (based on ideas from a similar claim in page 11 of [LST20]).

Elementary proof of Claim 2. From Claim 1, it is enough to show that

$$E_p(\mathbb{F}_p)[\ell^\infty] \cong T_\ell E_p / (1 - F_p) T_\ell E_p$$

Denote $f := 1 - F_p : T_\ell E_p \rightarrow T_\ell E_p$. $T_\ell E_p \cong \mathbb{Z}_\ell^2$ and by the structure theorem for finitely generated abelian groups/modules over PIDs we can choose bases such that

$$f = \begin{pmatrix} \ell^a & 0 \\ 0 & \ell^b \end{pmatrix}$$

where $a \leq b$. From this we get

$$T_\ell E_p / f T_\ell E_p \cong \mathbb{Z} / \ell^a \mathbb{Z} \times \mathbb{Z} / \ell^b \mathbb{Z}$$

Taking n larger enough so that $E_p(\mathbb{F}_p)[\ell^n] = E_p(\mathbb{F}_p)[\ell^\infty]$ and $a, b \leq n$ we get that f_n , the projection of f on $E_p(\mathbb{F}_p)[\ell^n]$, is also represented by $\begin{pmatrix} \ell^a & 0 \\ 0 & \ell^b \end{pmatrix}$ as a matrix in $M_2(\mathbb{Z} / \ell^n \mathbb{Z})$. f_n 's kernel is by definition $E_p(\mathbb{F}_p)[\ell^n] = E_p(\mathbb{F}_p)[\ell^\infty]$. On the other hand under the isomorphism given by picking bases we see that f_n 's kernel is $\left\{ \begin{pmatrix} \ell^{n-a} x \\ \ell^{n-b} y \end{pmatrix} \in (\mathbb{Z} / \ell^n \mathbb{Z})^2 \mid x, y \in \mathbb{Z} \right\}$ which is isomorphic to $\mathbb{Z} / \ell^a \mathbb{Z} \times \mathbb{Z} / \ell^b \mathbb{Z}$ via the map $(e_1, e_2) \mapsto \begin{pmatrix} \ell^{n-a} e_1 \\ \ell^{n-b} e_2 \end{pmatrix}$. And so we get

$$E(\mathbb{F}_p)[\ell^\infty] \cong T_\ell E_p / (1 - F_p) T_\ell E_p$$

$\square$

Much nicer proof of Claim 2. From Claim 1, it is enough to show that

$$E_p(\mathbb{F}_p)[\ell^\infty] \cong T_\ell E_p / (1 - F_p) T_\ell E_p$$

Looking at the following commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & T_\ell E_p & \longrightarrow & T_\ell E_p \otimes \mathbb{Q}_\ell & \longrightarrow & T_\ell E_p \otimes \mathbb{Q}_\ell / T_\ell E_p \longrightarrow 0 \\ & & \downarrow 1-F_p & & \downarrow 1-F_p & & \downarrow 1-F_p \\ 0 & \longrightarrow & T_\ell E_p & \longrightarrow & T_\ell E_p \otimes \mathbb{Q}_\ell & \longrightarrow & T_\ell E_p \otimes \mathbb{Q}_\ell / T_\ell E_p \longrightarrow 0 \end{array}$$

$T_\ell E_p \otimes \mathbb{Q}_\ell$ is a finite dimensional vector space over $\mathbb{Q}_\ell$ on which $1 - F_p$ is injective and hence also surjective, so the snake lemma gives us an exact sequence

$$0 \longrightarrow \ker \left(1 - F_p \Big| T_\ell E_p \otimes \mathbb{Q}_\ell / T_\ell E_p \right) \longrightarrow \text{coker}(1 - F_p \mid T_\ell E_p) \longrightarrow 0$$

where the cokernel is the right hand side of the isomorphism we are trying to prove. Now note that $T_\ell E_p \otimes \mathbb{Q}_\ell / T_\ell E_p$ is isomorphic to $E_p(\overline{\mathbb{F}_p})[\ell^\infty]$, because we can look at the map $f : T_\ell E_p \otimes \mathbb{Q}_\ell \rightarrow E_p(\overline{\mathbb{F}_p})[\ell^\infty]$ which sends any basic tensor written as $(P_i)_{i=1}^\infty \otimes \frac{1}{i^n}$ (where n is minimal, i.e $P_1 \neq 0$) to P_n . This map is surjective and its kernel is exactly $T_\ell E_p$ (naturally embedded in the tensor product), so it induces an isomorphism on the quotient. Following the action of F_p , we see that almost tautologically it is sent to himself, so $\ker(1 - F_p \mid T_\ell E_p \otimes \mathbb{Q}_\ell / T_\ell E_p) = \ker(1 - F_p \mid E(\overline{\mathbb{F}_p})[\ell^\infty]) = E(\mathbb{F}_p)[\ell^\infty]$ and we finished. $\square$

Now we arrive at the main theorem of this section:

Theorem 4. *Given an elliptic curve E over $\mathbb{Q}$ without CM, for every abelian ℓ -group G and every subset $C \subseteq \mathbb{N}$:*

$$\mathbb{P}(p \in C, E(\mathbb{F}_p)[\ell^\infty] \cong G) = [GL_2(\mathbb{Z}_\ell) : \text{im}(\rho_\ell)] \cdot \mu \left(\left\{ M \in \text{im}(\rho_\ell) \mid \det(M) \in C, \mathbb{Z}_\ell^2 / (1 - M) \mathbb{Z}_\ell^2 \cong G \right\} \right)$$

Where μ is the probability Haar measure on $GL_2(\mathbb{Z}_\ell)$ and $\rho_\ell : G_\mathbb{Q} \rightarrow GL_2(\mathbb{Z}_\ell)$ is the ℓ -adic representation attached to E .

Proof. Firstly, from Claim 2 and the fact that $\det(Frob_p) = p$ we get that

$$\mathbb{P}(p \in C \text{ and } E(\mathbb{F}_p)[\ell^\infty] \cong G) = \mathbb{P}(\det(Frob_p) \in C \text{ and } T_\ell E / (1 - Frob_p) T_\ell E \cong G)$$

By Theorem 2, the image of ρ_ℓ in $GL_2(\mathbb{Z}_\ell)$ is of finite index, so $\text{im}(\rho_\ell) \cong \text{Gal}(K/\mathbb{Q})$ for some

extension $K/\mathbb{Q}$ and we get the following diagram:

$$\begin{array}{ccc} G_{\mathbb{Q}} & \xrightarrow{\rho_{\ell}} & GL_2(\mathbb{Z}_{\ell}) \\ \downarrow & \nearrow & \\ \text{Gal}(K/\mathbb{Q}) & & \end{array}$$

Note that from this diagram, we get that $K/\mathbb{Q}$ is unramified for all but finitely many primes. Specifically, E has good reduction for all but finitely many primes and for every $p \neq \ell$ such that E has good reduction at p , $K/\mathbb{Q}$ is unramified at p : By the Neron-Ogg-Shafarevich criterion (Theorem 1) we have $\rho_{\ell}(I_p) = 0$ but then from injectivity of $\text{Gal}(K/\mathbb{Q}) \hookrightarrow GL_2(\mathbb{Z}_{\ell})$ we get that the image of I_p in $\text{Gal}(K/\mathbb{Q})$ is also 0, meaning $K/\mathbb{Q}$ is unramified at p . So now we can use Chebotarev's density theorem (Theorem 3) and after noticing that $[GL_2(\mathbb{Z}_{\ell}) : \text{im}(\rho_{\ell})] \cdot \mu$ is the probability Haar measure on $\text{im}(\rho_{\ell})$ we get:

$$\begin{aligned} & \mathbb{P}\left(\det(Frob_p) \in C \text{ and } T_{\ell}E/(1 - Frob_p)T_{\ell}E \cong G\right) = \\ & = [GL_2(\mathbb{Z}_{\ell}) : \text{im}(\rho_{\ell})] \cdot \mu\left(\left\{M \in \text{im}(\rho_{\ell}) \mid \det(M) \in C \text{ and } \mathbb{Z}_{\ell}^2/(1 - M)\mathbb{Z}_{\ell}^2 \cong G\right\}\right) \end{aligned}$$

□

Now we want to continue and make this probability more explicit. The first step is to note that by Theorem 2, ρ_{ℓ} is surjective for all but finitely many ℓ , and so adding the assumption that ρ_{ℓ} is surjective is not so restrictive. Furthermore, $E(\mathbb{F}_p)[\ell^{\infty}]$ is an abelian ℓ -group of rank ≤ 2 , and so we are interested in taking G to be such a group. In addition, the following claim convinces it is reasonable to restrict to primes p such that $v_{\ell}(p - 1)$ is fixed.

Claim 3. *For all $M \in GL_2(\mathbb{Z}_{\ell})$, if we write $\text{coker}(1 - M) = \mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z}$ where $e_1 \leq e_2$ then $e_1 \leq v_{\ell}(1 - \det(M))$.*

Proof. Writing $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, by the structure theorem for finitely generated modules over PIDs $\gcd(1 - a, b, c, 1 - d) = (\ell^{e_1})$ so it is enough to show that $1 - \det(M) \in \gcd(1 - a, b, c, 1 - d)$ and this is true because

$$1 - \det(M) = 1 - ad + bc = -(1 - a)(1 - d) + bc + (1 - a) + (1 - d)$$

and so $1 - \det(M) \in \gcd(1 - a, b, c, 1 - d)$ and $e_1 = v_{\ell}(\ell^{e_1}) \leq v_{\ell}(1 - \det(M))$

□

Now we arrive at the following claim, which is a more explicit version of Theorem 4.

Claim 4. Fix some prime number ℓ and $t \in \mathbb{Z}_{\geq 0}$. For every pair of integers $0 \leq e_1 \leq e_2$, assuming ρ_ℓ is surjective:

$$\mathbb{P}\left(v_\ell(p-1) = t, E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z}\right) = \mu\left(\left\{M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathbb{Z}_\ell) \left| \begin{array}{l} v_\ell(\det(M) - 1) = t, \\ v_\ell(\det(1 - M)) = e_1 + e_2, \\ \gcd(1 - a, b, c, 1 - d) = \ell^{e_1} \end{array} \right. \right\}\right)$$

Proof. Denote $G := \mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z}$. By Theorem 4 we have:

$$\mathbb{P}(v_\ell(p-1) = t, E(\mathbb{F}_p)[\ell^\infty] \cong G) = \mu\left(\left\{M \in GL_2(\mathbb{Z}_\ell) \left| v_\ell(\det(M) - 1) = t, \mathbb{Z}_\ell^2/(1 - M)\mathbb{Z}_\ell^2 \cong G \right.\right\}\right)$$

And by the structure theorem for finitely generated modules over PIDs $\mathbb{Z}_\ell^2/(1 - M)\mathbb{Z}_\ell^2 \cong G$ if and only if $1 - M$'s smith normal form is $\begin{pmatrix} \ell^{e_1} & 0 \\ 0 & \ell^{e_2} \end{pmatrix}$, meaning the gcd of $1 - M$'s coefficients is ℓ^{e_1} and its determinant is $\ell^{e_1+e_2}$ and so we get the wanted result. $\square$

Remark. It is interesting to note that $\det(Frob_p) = p$ and so for $p \not\equiv 1 \pmod{\ell}$, Claim 2 and Claim 3 imply that $E(\mathbb{F}_p)[\ell^\infty]$ is cyclic. This is a claim that can also be proved using the Weil pairing.

Remark. Stated differently, Claim 3 implies that if we fix $v_\ell(1 - \det(M)) =: t$ then the only possible groups for $\text{coker}(1 - M)$ are $\mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z}$ where $e_1 \leq t$ and $e_1 \leq e_2$. For example, if we restrict to $v_\ell(1 - \det(M)) = 0$ then $\text{coker}(1 - M)$ is necessarily cyclic. This is a part of our strategy to compute specific cases in Section 4.

4 Computational Results

In this section we continue from Section 3 and prove the following theorem:

Theorem 5. *Let E be an elliptic curve over $\mathbb{Q}$ without CM and ℓ a prime number.*

$$\begin{aligned} \mathbb{P}(E(\mathbb{F}_p)[\ell^\infty] = 0) &= 1 - \frac{\ell^2}{(\ell-1)^2(\ell+1)} \\ \mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e\mathbb{Z} \mid v_\ell(p-1) = 0\right) &= \begin{cases} \frac{\ell-2}{\ell-1} & e = 0 \\ \frac{1}{\ell^e} & e \geq 1 \end{cases} \\ \mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e\mathbb{Z} \mid v_\ell(p-1) = 1\right) &= \begin{cases} 1 - \frac{\ell}{\ell^2-1} & e = 0 \\ \frac{\ell-1}{\ell^{e+1}} & e \geq 1 \end{cases} \\ \mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e\mathbb{Z} \times \mathbb{Z}/\ell\mathbb{Z} \mid v_\ell(p-1) = 1\right) &= \begin{cases} \frac{\ell^2-\ell-1}{\ell^3(\ell^2-1)} & e = 1 \\ \frac{1}{\ell^{e+2}} & e \geq 2 \end{cases} \\ \mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e\mathbb{Z} \mid v_\ell(p-1) = 2\right) &= \begin{cases} \frac{\ell-1}{\ell^e} & e \geq 3 \end{cases} \end{aligned}$$

From Claim 3 and the remarks after it, these cases conclude the cases $v_\ell(p-1) = 0$ and $v_\ell(p-1) = 1$.

4.1 Notations

throughout this section we fix an elliptic curve E over $\mathbb{Q}$ without CM and ℓ a prime number such that ρ_ℓ is surjective.

For every $k \geq 1$, define $\mathcal{O}_k := \mathbb{Z}_\ell / \ell^k \mathbb{Z}_\ell \cong \mathbb{Z} / \ell^k \mathbb{Z}$. On $\mathcal{O}_k$ we define the reduced "valuation" function $v : \mathcal{O}_k \rightarrow \{0, \dots, k\}$ by $\pi^{-1}((a)) = (\ell^{v(a)})$ for all $a \in \mathcal{O}_k$ where $\pi : \mathbb{Z}_\ell \rightarrow \mathcal{O}_k$ is the projection map. In other words, if we take a representative $x \in \mathbb{Z}_\ell$ of some coset in $\mathcal{O}_k$ then if $v_\ell(x) < k$ we take $v(\bar{x}) = v_\ell(x)$ and if $v_\ell(x) \geq k$ we take $v(\bar{x}) = k$ (note that this does not depend on choice of representative). each element $a \in \mathcal{O}_k$ can be written uniquely as a sum $a_0 + a_1\ell + a_2\ell^2 + \dots + a_{k-1}\ell^{k-1}$ where $a_0, \dots, a_{k-1} \in \{0, \dots, \ell-1\}$ and $v(a) \geq t$ if and only if $a_0 = \dots = a_{t-1} = 0$.

4.2 Plan

In each subsection in this section, we calculate a different case of the distribution of $E(\mathbb{F}_p)[\ell^\infty]$. We start by calculating the probability that $E(\mathbb{F}_p)[\ell^\infty]$ is trivial. Afterwards we compute probabilities conditional on fixing $v_\ell(p-1)$. After treating the first case separately, we develop a plan to calculate each case where $v_\ell(p-1)$ is fixed and we use it in each case. The main steps are to use the ideas of Section 2.2 to reduce the question to a counting problem about matrices over a finite ring and then solve that problem using an inclusion-exclusion formula (see for example Claim 8).

4.3 Trivial group, $E(\mathbb{F}_p)[\ell^\infty] = 0$

This subsection is stand-alone, without the use of results from Section 3.

Claim 5. *Let $M \in GL_2(\mathbb{F}_\ell)$ be a uniformly chosen matrix. We have*

$$\mathbb{P}(E(\mathbb{F}_p)[\ell^\infty] = 0) = \mathbb{P}(M \text{ has no non-trivial fixed points})$$

Where the probability on the left is actually the natural density of primes p such that $E(\mathbb{F}_p)[\ell^\infty] = 0$.

Proof. By projecting from $\mathbb{Z}_\ell$ to $\mathbb{F}_\ell$ we get $\overline{\rho}_\ell : G_\mathbb{Q} \rightarrow GL_2(\mathbb{F}_\ell)$ which is still surjective. $GL_2(\mathbb{F}_\ell)$ is finite so $\ker(\overline{\rho}_\ell)$ has finite index in $G_\mathbb{Q}$ and hence denoting by $K \subseteq \overline{\mathbb{Q}}$ the finite extension of $\mathbb{Q}$ fixed by $\ker(\overline{\rho}_\ell)$ we get the following commutative diagram:

$$\begin{array}{ccc} G_\mathbb{Q} & \twoheadrightarrow & GL_2(\mathbb{F}_\ell) \\ & \searrow & \uparrow \wr \\ & & \text{Gal}(K/\mathbb{Q}) \end{array}$$

Under the isomorphism $\text{Gal}(K/\mathbb{Q}) \xrightarrow{\sim} GL_2(\mathbb{F}_\ell)$, $\overline{\rho}_\ell(\text{Frob}_p)$ corresponds to "the" (up to conjugation) p -th power Frobenius element in $\text{Gal}(K/\mathbb{Q})$. Now note that

$$\begin{aligned} E(\mathbb{F}_p)[\ell^\infty] = 0 &\Leftrightarrow \ker(1 - F_p | E(\overline{\mathbb{F}_p})[\ell]) = E(\mathbb{F}_p)[\ell] = 0 \Leftrightarrow \\ &\Leftrightarrow \ker(1 - \overline{\rho}_\ell(\text{Frob}_p)) = 0 \Leftrightarrow \overline{\rho}_\ell(\text{Frob}_p) \text{ has no non-trivial fixed points} \end{aligned}$$

Denote by A the set of matrices in $GL_2(\mathbb{F}_\ell)$ that don't have non-trivial fixed points. A is invariant under conjugation and so by Chebotarev's density theorem (even the classical finite extension version):

$$\begin{aligned} \mathbb{P}(M \text{ has no non-trivial fixed points}) &= \mathbb{P}(M \in A) = \\ &= \lim_{X \rightarrow \infty} \frac{\#\{p < X \mid \overline{\rho}_\ell(\text{Frob}_p) \text{ has no non-trivial fixed points}\}}{\#\{p < X\}} = \\ &= \lim_{X \rightarrow \infty} \frac{\#\{p < X \mid E(\mathbb{F}_p)[\ell^\infty] = 0\}}{\#\{p < X\}} = \mathbb{P}(E(\mathbb{F}_p)[\ell^\infty] = 0) \end{aligned}$$

□

Claim 6. Let $M \in GL_2(\mathbb{F}_\ell)$ be a uniformly chosen matrix, then:

$$\mathbb{P}(M \text{ has no fixed points}) = 1 - \frac{\ell^2}{(\ell-1)^2(\ell+1)}$$

Proof. Firstly note that $\#GL_2(\mathbb{F}_\ell) = (\ell^2 - 1)(\ell^2 - \ell)$. Now we need to count how many matrices in $GL_2(\mathbb{F}_\ell)$ don't have any fixed points, i.e 1 isn't an eigenvalue. We'll do the opposite and count the matrices such that 1 is an eigenvalue. The degree of the characteristic polynomial is two and has a root so it splits completely and the matrix has a Jordan form. now we'll go through all the Jordan forms with eigenvalue 1 and calculate their conjugacy class. From uniqueness of Jordan form the sum of the sizes of the conjugacy classes will be the number of matrices with eigenvalue 1.

$(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix})$: Start by calculating the size of the centralizer $C_{GL_2(\mathbb{F}_\ell)}((\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}))$. A matrix $(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix})$ is in $C_{GL_2(\mathbb{F}_\ell)}((\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}))$ if and only if:

$$\begin{aligned} & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ \Leftrightarrow & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} (ad - bc) \\ \Leftrightarrow & \begin{pmatrix} ad - ac - bc & a^2 \\ c^2 & ac + ad - bc \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} (ad - bc) \end{aligned}$$

which is equivalent to $a = d \wedge c = 0$ and so there are $\ell - 1$ options for a (0 is not an option because $ad - bc \neq 0$) and ℓ options for b , meaning that $\#C_{GL_2(\mathbb{F}_\ell)}((\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix})) = \ell(\ell - 1)$. By the orbit-stabilizer theorem we get that the size of the conjugacy class of $(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix})$ is $\frac{\#GL_2(\mathbb{F}_\ell)}{\#C_{GL_2(\mathbb{F}_\ell)}((\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}))} = \frac{(\ell^2 - 1)(\ell^2 - \ell)}{\ell(\ell - 1)} = \ell^2 - 1$.

$(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix})$: Scalar matrices commute with every other matrix so its conjugacy class has size 1.

$(\begin{smallmatrix} 1 & 0 \\ 0 & \lambda \end{smallmatrix})$, $\lambda \notin \{0, 1\}$: As in the first case, we calculate the size of the centralizer $C_{GL_2(\mathbb{F}_\ell)}((\begin{smallmatrix} 1 & 0 \\ 0 & \lambda \end{smallmatrix}))$. A matrix $(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix})$ is in $C_{GL_2(\mathbb{F}_\ell)}((\begin{smallmatrix} 1 & 0 \\ 0 & \lambda \end{smallmatrix}))$ if and only if:

$$\begin{aligned} & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} \\ \Leftrightarrow & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} (ad - bc) \\ \Leftrightarrow & \begin{pmatrix} ad - \lambda bc & ab(\lambda - 1) \\ cd(1 - \lambda) & -bc + \lambda ad \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} (ad - bc) \end{aligned}$$

which is equivalent to $a \neq 0 \wedge d \neq 0 \wedge b = 0 \wedge c = 0$, meaning $\#C_{GL_2(\mathbb{F}_\ell)}((\begin{smallmatrix} 1 & 0 \\ 0 & \lambda \end{smallmatrix})) = (\ell - 1)^2$ and the

size of the conjugacy class of $\begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix}$ is $\frac{\#GL_2(\mathbb{F}_\ell)}{\#C_{GL_2(\mathbb{F}_\ell)}(\begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix})} = \frac{(\ell^2-1)(\ell^2-\ell)}{(\ell-1)^2} = \ell(\ell+1)$.

Now we sum and find that the number of matrices in $GL_2(\mathbb{F}_\ell)$ without fixed points is

$$\begin{aligned} & (\ell^2 - 1)(\ell^2 - \ell) - [(\ell^2 - 1) + 1 + (\ell - 2)\ell(\ell + 1)] = \ell(\ell^3 - 2\ell^2 - \ell + 1) \\ \Rightarrow \quad \mathbb{P}(M \text{ has no fixed points}) &= \frac{\ell(\ell^3 - 2\ell^2 - \ell + 1)}{(\ell^2 - 1)(\ell^2 - \ell)} = 1 - \frac{\ell^2}{(\ell - 1)^2(\ell + 1)} \end{aligned}$$

□

4.4 Cyclic case, $G = \mathbb{Z}/\ell^e\mathbb{Z}$, $v_\ell(p - 1) = 0$

We define the following quantities:

$$N(k, D) := \#\{M \in GL_2(\mathcal{O}_k) \mid \det(1 - M) = 0 \text{ and } \det(M) = D\}$$

$$M(k, D) := \#\{M \in GL_2(\mathcal{O}_k) \mid v(\det(1 - M)) = k - 1 \text{ and } \det(M) = D\}$$

$$M(k) := \#\{M \in GL_2(\mathcal{O}_k) \mid v(\det(1 - M)) = k - 1 \text{ and } v(\det(M) - 1) = 0\} = \sum_{D \in \mathcal{O}_k^\times, v(D-1)=0} M(k, D)$$

Claim 7. For every $k \geq 1$ and $D \in \mathcal{O}_k^\times$ such that $v(D - 1) = 0$:

$$N(k, D) = \ell^{2k-1}(\ell + 1)$$

Proof. Writing $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We are counting all matrices such that $ad - bc = D$ and $1 - a - d + D = 0$. For every a , d is uniquely determined by $d = 1 - a + D$, so we only need to count options for a, b, c . Furthermore, $bc = ad - D = a(1 - a + D) - D = (1 - a)(a - D)$, so $v(b) \leq v((1 - a)(a - D))$ and for every b there are $\ell^{v(b)}$ options for c (lemma 15 in [MS21]). Lastly, note that $1 - D = (1 - a) + (a - D)$ so only one of $1 - a$, $a - D$ can have positive valuation. Altogether we get:

$$\begin{aligned} N(k, D) &= \sum_{i=1}^k \#\{a \in \mathcal{O}_k \mid v(1 - a) = i\} \sum_{j=0}^i \#\{b \in \mathcal{O}_k \mid v(b) = j\} \ell^j + \\ &+ \sum_{i=1}^k \#\{a \in \mathcal{O}_k \mid v(a - D) = i\} \sum_{j=0}^i \#\{b \in \mathcal{O}_k \mid v(b) = j\} \ell^j + \ell^{2k-2}(\ell - 1)(\ell - 2) \end{aligned}$$

Simple counting reveals that $\#\{\gamma \in \mathcal{O}_k \mid v(\gamma) = j\}$ equals $\ell^{k-1-j}(\ell-1)$ if $j < k$ and 1 if $j = k$, so:

$$\begin{aligned}
N(k, D) &= 2 \left(\ell^k + k\ell^{k-1}(\ell-1) + \sum_{i=1}^{k-1} \ell^{k-1-i}(\ell-1)(i+1)\ell^{k-1}(\ell-1) \right) + \ell^{2k-2}(\ell-1)(\ell-2) = \\
&= 2(k+1)\ell^k - 2k\ell^{k-1} + (\ell-1)(\ell-2)\ell^{2k-2} + 2(\ell-1)^2\ell^{2k-2} \sum_{i=1}^{k-1} \frac{i+1}{\ell^i} = \\
&\stackrel{(*)}{=} 2(k+1)\ell^k - 2k\ell^{k-1} + (\ell-1)(\ell-2)\ell^{2k-2} + 2\ell^{k-2}(2\ell^{k+1} - \ell^k - (k+1)\ell^2 + k\ell) = \\
&= 2(k+1)\ell^k - 2k\ell^{k-1} + (\ell-1)(\ell-2)\ell^{2k-2} + 4\ell^{2k-1} - 2\ell^{2k-2} - 2(k+1)\ell^k + 2k\ell^{k-1} = \\
&= (\ell^2 - 3\ell + 2)\ell^{2k-2} + 4\ell^{2k-1} - 2\ell^{2k-2} = \ell^{2k} + \ell^{2k-1} = \ell^{2k-1}(\ell+1)
\end{aligned}$$

Where the equality denoted by $(*)$ is true because $\sum_{i=1}^{k-1} \frac{i+1}{\ell^i} = \frac{2\ell^k - \ell^{k-1} - (k+1)\ell + k}{\ell^{k-1}(\ell-1)^2}$ □

Claim 8. For every $k \geq 2$ and $D \in \mathcal{O}_k^\times$ such that $v(D-1) = 0$, denoting by $\pi : \mathcal{O}_k \rightarrow \mathcal{O}_{k-1}$ the projection, we have:

$$M(k, D) = \ell^3 N(k-1, \pi(D)) - N(k, D)$$

Proof. Denote

$$\begin{aligned}
X_{D'} &:= \{M \in GL_2(\mathcal{O}_k) \mid \det(M) = D', v(\det(1-M)) \geq k-1\}, \forall D' \in \pi^{-1}(\pi(D)) \\
Y &:= \{M \in GL_2(\mathcal{O}_{k-1}) \mid \det(M) = \pi(D), \det(1-M) = 0\}
\end{aligned}$$

and note that $\#X_D = M(k, D) + N(k, D)$, $\#Y = N(k-1, \pi(D))$ and furthermore $\#X_{D_1} = \#X_{D_2}$ for all $D_1, D_2 \in \pi^{-1}(\pi(D))$, because multiplying the top row by $\frac{D_2}{D_1}$ is a bijection $X_{D_1} \rightarrow X_{D_2}$. The projection $\pi : \mathcal{O}_k \rightarrow \mathcal{O}_{k-1}$ induces a map $GL_2(\mathcal{O}_k) \rightarrow GL_2(\mathcal{O}_{k-1})$ that restricts to a surjective map $\bigcup_{D' \in \pi^{-1}(\pi(D))} X_{D'} \twoheadrightarrow Y$ where each fiber has size $C := \#\ker(GL_2(\mathcal{O}_k) \rightarrow GL_2(\mathcal{O}_{k-1}))$, so:

$$\begin{aligned}
\# \left(\bigcup_{D' \in \pi^{-1}(\pi(D))} X_{D'} \right) &= C \cdot \#Y \\
\underbrace{\#\pi^{-1}(\pi(D))}_{=\ell} \cdot \#X_D &= \sum_{D' \in \pi^{-1}(\pi(D))} \#X_{D'} = C \cdot \#Y \\
\Rightarrow M(k, D) + N(k, D) &= \frac{C}{\ell} \cdot N(k-1, \pi(D))
\end{aligned}$$

It remains to show that $C = \ell^4$, which is true because

$$\begin{aligned} \ker(GL_2(\mathcal{O}_k) \rightarrow GL_2(\mathcal{O}_{k-1})) &= 1 + \ell^{k-1}M_2(\mathcal{O}_k) = 1 + M_2(\ell^{k-1}\mathcal{O}_k) \\ \Rightarrow C = \#M_2(\ell^{k-1}\mathcal{O}_k) &= (\ell^{k-1}\mathcal{O}_k)^4 = \ell^4 \end{aligned}$$

The last step is true because $\ell^{k-1}\mathcal{O}_k$ is the set of elements of $\mathcal{O}_k$ with reduced valuation $\geq k-1$ $\square$

Corollary 1. *For every $k \geq 2$ and $D \in \mathcal{O}_k^\times$ such that $v(D-1) = 0$*

$$M(k, D) = \ell^3 N(k-1, \pi(D)) - N(k, D) = \ell^3 \ell^{2k-3}(\ell+1) - \ell^{2k-1}(\ell+1) = \ell^{2k-1}(\ell+1)(\ell-1)$$

Corollary 2. *For every $k \geq 2$*

$$\begin{aligned} M(k) &= \sum_{D \in \mathcal{O}_k^\times, v(D-1)=0} M(k, D) = \\ &= \underbrace{\#\{D \in \mathcal{O}_k^\times, v(D-1)=0\}}_{=\ell^{k-1}(\ell-2)} \cdot \ell^{2k-1}(\ell+1)(\ell-1) = \ell^{3k-2}(\ell+1)(\ell-1)(\ell-2) \end{aligned}$$

We calculated $M(k)$ for $k \geq 2$, but for our last claim we also want $M(1)$.

Claim 9.

$$M(1) = (\ell-2)^2 \ell(\ell+1)$$

Proof. $M(1)$ exactly counts matrices in $GL_2(\mathbb{F}_\ell) \setminus SL_2(\mathbb{F}_\ell)$ for which 1 is not an eigenvalue. In $GL_2(\mathbb{F}_\ell) \setminus SL_2(\mathbb{F}_\ell)$, having 1 as an eigenvalue is equivalent to having Jordan form $\begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix}$ for $\lambda \notin \{0, 1\}$ and from the calculations in the proof of Claim 6 we know there are $\ell(\ell+1)$ such matrices for each λ , so

$$\begin{aligned} M(1) &= \#GL_2(\mathbb{F}_\ell) \setminus SL_2(\mathbb{F}_\ell) - (\ell-2)\ell(\ell+1) = ((\ell^2-1)(\ell^2-\ell) - (\ell^2-1)\ell) - (\ell-2)\ell(\ell+1) = \\ &= (\ell^2-1)\ell(\ell-2) - (\ell-2)\ell(\ell+1) = (\ell-2)\ell(\ell^2-\ell-2) = (\ell-2)^2 \ell(\ell+1) \end{aligned}$$

$\square$

Claim 10. *For every $e \geq 0$*

$$\mu(\{M \in GL_2(\mathbb{Z}_\ell) \mid \det(M) \not\equiv 1 \pmod{\ell}, v_\ell(\det(1-M)) = e\}) = \begin{cases} \frac{(\ell-2)^2}{(\ell-1)^2} & e = 0 \\ \frac{\ell-2}{\ell^e(\ell-1)} & e \geq 1 \end{cases}$$

Proof. Taking $k = e + 1$, the set $X := \{M \in GL_2(\mathbb{Z}_\ell) \mid \det(M) \not\equiv 1 \pmod{\ell}, v_\ell(\det(1 - M)) = e\}$ is "detectable modulo ℓ^k " in the sense that $M \in X \Leftrightarrow \overline{M} \in \overline{X}$, where $\overline{X} \subseteq GL_2(\mathcal{O}_k)$ is the projection of X . As explained in Section 2.2, this gives us a decomposition of X as a disjoint union of cosets in $GL_2(\mathbb{Z}_\ell)/1 + \ell^{e+1}M_2(\mathbb{Z}_\ell) \cong GL_2(\mathcal{O}_{e+1})$ and those cosets are exactly the ones that $M(e + 1)$ is counting, so:

$$\mu(X) = \frac{M(e + 1)}{\#GL_2(\mathcal{O}_{e+1})}$$

Now to calculate the denominator, from the projection map $\pi : GL_2(\mathcal{O}_{e+1}) \rightarrow GL_2(\mathbb{F}_\ell)$ we get:

$$\#GL_2(\mathcal{O}_{e+1}) = \# \underbrace{\ker(\pi)}_{=1+\ell M_2(\mathcal{O}_{e+1})} \cdot \underbrace{\#GL_2(\mathbb{F}_\ell)}_{=(\ell^2-1)(\ell^2-\ell)} = \ell^{4e}(\ell^2 - 1)(\ell^2 - \ell)$$

And so for $e \geq 1$:

$$\mu(X) = \frac{M(e + 1)}{\#GL_2(\mathcal{O}_{e+1})} = \frac{\ell^{3e+1}(\ell + 1)(\ell - 1)(\ell - 2)}{\ell^{4e}(\ell^2 - 1)(\ell^2 - \ell)} = \frac{\ell - 2}{\ell^e(\ell - 1)}$$

And for $e = 0$:

$$\mu(X) = \frac{M(1)}{\#GL_2(\mathbb{F}_\ell)} = \frac{(\ell - 2)^2 \ell (\ell + 1)}{(\ell^2 - 1)(\ell^2 - \ell)} = \frac{(\ell - 2)^2}{(\ell - 1)^2}$$

□

Corollary 3. *The above claim concludes neatly the result that*

$$\mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e\mathbb{Z} \mid v_\ell(p - 1) = 0\right) = \begin{cases} \frac{\ell-2}{\ell-1} & e = 0 \\ \frac{1}{\ell^e} & e \geq 1 \end{cases}$$

given that ρ_ℓ is surjective.

Proof. Dirichlet's theorem on arithmetic progressions implies that the density of primes satisfying $v_\ell(p - 1) = 0$ is $\frac{\ell-2}{\ell-1}$. Dividing the joint probability we calculated before by this will give us the conditional probability we want. □

4.5 Cyclic case, $G = \mathbb{Z}/\ell^e\mathbb{Z}$, $v_\ell(p - 1) = 1$

We follow a similar strategy to 3.2, but cyclicity is not automatic.

As before, to calculate the probability for $\mathbb{Z}/\ell^e\mathbb{Z}$ we want to look at $GL_2(\mathcal{O}_{e+1})$, but now we are talking about $D \in \mathcal{O}_{e+1}$ such that $v(1 - D) = 1$ and we want to avoid talking about the "top"

valuation when using the reduced valuation, so we will now deal with $e \leq 1$ and afterwards deal with the general case $e \geq 2$.

For $k \leq 1$, define $M(k) = \#\{M \in GL_2(\mathcal{O}_2) \mid v(\det(M) - 1) = 1, v(\det(1 - M)) = k\}$ by the same reasoning as in Section 4.4, we have for $e \leq 1$:

$$\mathbb{P}\left(v_\ell(p-1) = 1, E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e\mathbb{Z}\right) = \frac{M(e)}{\#GL_2(\mathcal{O}_2)} = \frac{M(e)}{\ell^4(\ell^2-1)(\ell^2-\ell)}$$

Because for the trivial group and $\mathbb{Z}/\ell\mathbb{Z}$ it is enough to check the size.

So let's calculate $M(0)$. $A := \{M \in GL_2(\mathbb{F}_\ell) \mid \det(M) = 1, \det(1 - M) = 0\}$. From the calculations in the proof of Claim 6 through the Jordan form and orbit-stabilizer theorem, we know that $\#A = \ell^2$. So $\#SL_2(\mathbb{F}_\ell) \setminus A = \#\{M \in GL_2(\mathbb{F}_\ell) \mid \det(M) = 1, \det(1 - M) \neq 0\} = \ell(\ell^2 - 1) - \ell^2 = \ell(\ell^2 - \ell - 1)$. Note that $M(0)$ counts exactly the lifts of elements in $SL_2(\mathbb{F}_\ell) \setminus A$ that have determinant $\neq 1$.

Lemma 2. *For each $M \in GL_2(\mathbb{F}_\ell)$ such that $\det(M) = 1$, the lifts to $GL_2(\mathcal{O}_2)$ that do have determinant 1 are exactly a fiber of a non-trivial functional $\mathbb{F}_\ell^4 \rightarrow \mathbb{F}_\ell$*

Proof. Fix a lift $M_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of M and write $\det(M_0) = 1 + \ell t$ for some t . Every lift of M is of the form $M_0 + \ell M_1$ for some $M_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \in M_2(\mathcal{O}_2)$ and is uniquely determined by the projection of $M_1 \bmod \ell$.

$$\begin{aligned} \det(M_0 + \ell M_1) &= \det \begin{pmatrix} a + \ell a_1 & b + \ell b_1 \\ c + \ell c_1 & d + \ell d_1 \end{pmatrix} = (a + \ell a_1)(d + \ell d_1) - (b + \ell b_1)(c + \ell c_1) = \\ &= \underbrace{ad - bc}_{=\det(M)=1+\ell t} + \ell \underbrace{(a_1 d - b_1 c - c_1 b + a d_1)}_{=: f_1(a_1, b_1, c_1, d_1)} \end{aligned}$$

so the determinant of the lift is 1 if and only if $f_1(a_1, b_1, c_1, d_1) = -t$ which is a linear functional depending on the fixed M_0 . Note that this functional is non-trivial because $M_0 \neq 0$ $\square$

The size of such fiber is the size of the kernel, which is ℓ^3 so $\ell^4 - \ell^3$ lifts do contribute to $M(0)$, meaning that $M(0) = (\ell^4 - \ell^3)\ell(\ell^2 - \ell - 1) = \ell^4(\ell - 1)(\ell^2 - \ell - 1)$ and finally:

$$\mathbb{P}(v_\ell(p-1) = 1, E(\mathbb{F}_p)[\ell^\infty] \cong 0) = \frac{M(0)}{\#GL_2(\mathcal{O}_2)} = \frac{\ell^4(\ell-1)(\ell^2-\ell-1)}{\ell^4(\ell^2-1)(\ell^2-\ell)} = \frac{\ell^2-\ell-1}{\ell(\ell-1)}$$

Now we calculate $M(1)$.

Lemma 3. *For each $M \in GL_2(\mathbb{F}_\ell)$ such that $\det(1 - M) = 0$ and $M \neq I_2$, the lifts M_0 to $GL_2(\mathcal{O}_2)$ such that $\det(1 - M_0) = 0$ are exactly a fiber of a non-trivial functional $\mathbb{F}_\ell^4 \rightarrow \mathbb{F}_\ell$*

Proof. Fix a lift $M_0 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of M and write $\det(1 - M_0) = \ell t$ for some t . Every other lift is of the form $M_0 + \ell M_1$ for some $M_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \in M_2(\mathcal{O}_2)$ and is uniquely determined by the projection of $M_1 \bmod \ell$. Similarly to the previous lemma, for the condition $\det(1 - (M_0 + \ell M_1)) \neq 0$ we get a different functional:

$$\begin{aligned} \det(1 - (M_0 + \ell M_1)) &= \det \begin{pmatrix} 1 - a - \ell a_1 & -b - \ell b_1 \\ -c - \ell c_1 & 1 - d - \ell d_1 \end{pmatrix} = (1 - a - \ell a_1)(1 - d - \ell d_1) - (b + \ell b_1)(c + \ell c_1) = \\ &= \underbrace{(1 - a)(1 - d) - bc}_{=\det(1 - M) = \ell t} + \ell \underbrace{(a_1(1 - d) - b_1 c - c_1 b + d_1(1 - a))}_{=: f_2(a_1, b_1, c_1, d_1)} \end{aligned}$$

And this functional is trivial only for $M_0 = I_2$. $\square$

Note that $M(1)$ exactly counts lifts $M_0 \in GL_2(\mathcal{O}_2)$ of elements in A such that $\det(M_0) \neq 1$ and $\det(1 - M_0) \neq 0$. From the two lemmas, we see that the identity matrix $I_2 \in A$ does not have any good lifts but every other of the $\ell^2 - 1$ elements of A have

$$\ell^4 - \ell^3 - \ell^3 + \ell^2 = \ell^2(\ell - 1)^2$$

good lifts, where we add ℓ^2 because the two functionals written in the proofs, for the same fixed lift are linearly independent. Thus $M(1) = \ell^2(\ell - 1)^2 \cdot (\ell^2 - 1)$ and so:

$$\mathbb{P}(v_\ell(p - 1) = 1, E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell\mathbb{Z}) = \frac{M(1)}{\#GL_2(\mathcal{O}_2)} = \frac{\ell^2(\ell - 1)^2(\ell^2 - 1)}{\ell^4(\ell^2 - 1)(\ell^2 - \ell)} = \frac{(\ell - 1)^2}{\ell^2(\ell^2 - \ell)} = \frac{\ell - 1}{\ell^3}$$

Now we treat the general case, $e \geq 2$. We define the following quantities for $k \geq 2$ and $D \in \mathcal{O}_k^\times$

$$\begin{aligned} N(k, D) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} \det(1 - M) = 0, \det(M) = D, \\ \gcd(1 - a, b, c, 1 - d) = 1 \end{array} \right. \right\} \\ M(k, D) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} v(\det(1 - M)) = k - 1, \det(M) = D, \\ \gcd(1 - a, b, c, 1 - d) = 1 \end{array} \right. \right\} \\ M(k) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} v(\det(1 - M)) = k - 1, \\ v(1 - \det(M)) = 1, \\ \gcd(1 - a, b, c, 1 - d) = 1 \end{array} \right. \right\} = \sum_{D \in \mathcal{O}_k^\times, v(1 - D) = 1} M(k, D) \end{aligned}$$

Claim 11. For any $k \geq 2$ and $D \in \mathcal{O}_k^\times$ such that $v(1 - D) = 1$

$$N(k, D) = \ell^{2k-1}(\ell - 1)$$

Proof. Writing $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We are counting all matrices such that $ad - bc = D$, $1 - a - d + D = 0$ and $\min\{v(1-a), v(b), v(c), v(1-d)\} = 0$. For every a , d is uniquely determined by $d = 1 - a + D$, so we only need to count options for a, b, c . Furthermore, $bc = ad - D = ad - (a + d - 1) = (1-a)(1-d)$, so $v(b) \leq v((1-a)(1-d))$ and for every b there are $\ell^{v(b)}$ options for c (lemma 15 in [MS21]). Lastly, note that $1 - D = (1 - a) + (1 - d)$ so we have to following four cases contributing to $N(k, D)$:

1. $v(1 - a) = v(1 - d) = 0$
2. $v(1 - a) = v(1 - d) = 1$
3. $v(1 - a) = 1, v(1 - d) \geq 2$
4. $v(1 - a) \geq 2, v(1 - d) = 1$

Now we count each case separately.

1. $v(1 - a) = 0$ so the gcd is 1 always. Furthermore $v(b) = 0$ and there is 1 option for c for any b . Because $1 - d = a - D$ and $v(1 - D) = 1$ then $\overline{D} = 1 \in \mathbb{Z}_\ell / \ell \mathbb{Z}_\ell$ and $\overline{a} \neq 1 \in \mathbb{Z}_\ell / \ell \mathbb{Z}_\ell$. So we have $\ell - 1$ options for the first coordinate of a (the projection to $\mathbb{Z}_\ell / \ell \mathbb{Z}_\ell$) and ℓ options for each of the other coordinates and in general the contribution of this case to $N(k, D)$ is:

$$\underbrace{(\ell - 1)\ell^{k-1}}_{\text{options for } a} \underbrace{(\ell - 1)\ell^{k-1}}_{\text{options for } b} = \ell^{2k-2}(\ell - 1)^2$$

2. $v(1 - a) = v(1 - d) = 1$, so from $bc = (1 - a)(1 - d)$ we get $v(bc) = 2$ and we need at least one of b, c to have valuation 0. So the options are either $v(b) = 0$ and $v(c) = 2$ or the opposite, from symmetric reasoning it is enough to multiply by 2 the count of the first option. Writing $a = a_0 + a_1\ell + \dots + a_{k-1}\ell^{k-1}$, the constraints $v(1 - a) = v(a - D) = v(1 - D) = 1$ translate into $a_0 = 1$ and $a_1 \neq 0, D_1$, thus a has $(\ell - 2)\ell^{k-2}$ options and assuming $v(b) = 0$, b has $(\ell - 1)\ell^{k-1}$ options so together the contribution to $N(k, D)$ is:

$$2 \underbrace{(\ell - 2)\ell^{k-2}}_{\text{options for } a} \underbrace{(\ell - 1)\ell^{k-1}}_{\text{options for } b} = 2\ell^{2k-3}(\ell - 1)(\ell - 2)$$

3. By symmetric reasoning $a \leftrightarrow d$ this case contributes the same as case 4 below.
4. Denote $t := v(1 - a) \geq 2$. First of all, $\min\{v(b), v(c)\} = 0$ and $bc = ad - D$ which isn't a unit so exactly one of b, c is a unit, assume for now that $v(b) = 0$ and we'll double the result. So there are $(\ell - 1)\ell^{k-1}$ options for b and for each one there is a single option for c . If $t = k$ then $a = 1$, from $1 - a - d + D = 0$ we get $d = D$, so given b, c there is one option for a and d .

If $t < k$, writing $a = a_0 + a_1\ell + \dots + a_{k-1}\ell^{k-1}$ the constraints translate to $a_0 = 1, a_1 = \dots = a_{t-1} = 0, a_t \neq 0$ and we get $v(1-d) = v(a-D) = 1$ automatically, so the number of options for a is $(\ell-1)\ell^{k-t-1}$. Together the contribution of this case is:

$$\begin{aligned} & \underbrace{2(\ell-1)\ell^{k-1}}_{\text{options for } b} \cdot \underbrace{\left(1 + \sum_{t=2}^{k-1} (\ell-1)\ell^{k-t-1}\right)}_{\text{options for } a} = 2(\ell-1)\ell^{k-1} \left(1 + (\ell-1) \sum_{j=0}^{k-3} \ell^j\right) = \\ & = 2(\ell-1)\ell^{k-1} (1 + \ell^{k-2} - 1) = 2\ell^{2k-3}(\ell-1) \end{aligned}$$

Now we sum all cases and finally get:

$$\begin{aligned} N(k, D) &= \ell^{2k-2}(\ell-1)^2 + 2\ell^{2k-3}(\ell-1)(\ell-2) + 2 \cdot 2\ell^{2k-3}(\ell-1) = \\ &= \ell^{2k-3}(\ell-1)(\ell(\ell-1) + 2(\ell-2) + 4) = \ell^{2k-2}(\ell-1)(\ell+1) = \ell^{2k-2}(\ell^2-1) \end{aligned}$$

□

Claim 12. Denoting by $\pi : \mathcal{O}_k \rightarrow \mathcal{O}_{k-1}$ the projection. For every $k \geq 3$ and $D \in \mathcal{O}_k^\times$ such that $v(1-D) = 1$:

$$M(k, D) = \ell^3 N(k-1, \pi(D)) - N(k, D)$$

Proof. Exactly the proof of Claim 8. We didn't assume anything on D there other than $D \in \mathcal{O}_k^\times$. Note that for $k \geq 3$ then two matrices $M_1, M_2 \in GL_2(\mathcal{O}_k)$ in the same fiber of the projection $GL_2(\mathcal{O}_k) \rightarrow GL_2(\mathcal{O}_{k-1})$ differ by a matrix multiple of ℓ^{k-1} and so a coefficient of $1 - M_1$ is divisible by ℓ if and only if the corresponding coefficient of $1 - M_2$ is divisible by ℓ . □

Corollary 4. For every $k \geq 3$ and $D \in \mathcal{O}_k^\times$ such that $v(1-D) = 1$

$$M(k, D) = \ell^3 N(k-1, \pi(D)) - N(k, D) = \ell^{2k-1}(\ell^2-1) - \ell^{2k-2}(\ell^2-1) = \ell^{2k-2}(\ell^2-1)(\ell-1)$$

Corollary 5. For every $k \geq 3$

$$\begin{aligned} M(k) &= \sum_{D \in \mathcal{O}_k^\times, v(1-D)=1} M(k, D) = \\ &= \underbrace{\#\{D \in \mathcal{O}_k^\times, v(1-D)=1\}}_{=\ell^{k-2}(\ell-1)} \cdot \ell^{2k-2}(\ell^2-1)(\ell-1) = \ell^{3k-4}(\ell^2-1)(\ell-1)^2 \end{aligned}$$

Claim 13. *For every $e \geq 2$*

$$\mu \left(\left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathbb{Z}_\ell) \mid \begin{array}{l} v_\ell(\det(M) - 1) = 1, \\ v_\ell(\det(1 - M)) = e, \\ \gcd(1 - a, b, c, 1 - d) = 1 \end{array} \right\} \right) = \frac{\ell - 1}{\ell^{e+2}}$$

Proof. As in the proof of claim 9. Denote this set by X . So X is "detectable modulo ℓ^k " for $k = e + 1$ and we get

$$\mu(X) = \frac{M(e+1)}{\#GL_2(\mathcal{O}_{e+1})} = \frac{\ell^{3e-1}(\ell^2 - 1)(\ell - 1)^2}{\ell^{4e}(\ell^2 - 1)(\ell^2 - \ell)} = \frac{\ell - 1}{\ell^{e+2}}$$

□

Corollary 6. *The above claim concludes neatly the result that for all $e \geq 0$*

$$\mathbb{P} \left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e \mathbb{Z} \mid v_\ell(p - 1) = 1 \right) = \begin{cases} 1 - \frac{\ell}{\ell^2 - 1} & e = 0 \\ \frac{\ell - 1}{\ell^{e+1}} & e \geq 1 \end{cases}$$

given that ρ_ℓ is surjective.

Proof. Dirichlet's theorem on arithmetic progressions implies that the density of primes satisfying $v_\ell(p - 1) = 1$ is $\frac{1}{\ell}$. Dividing the joint probabilities we calculated before by this will give us the conditional probabilities we want. □

4.6 Non-cyclic case, $G = \mathbb{Z}/\ell^e \mathbb{Z} \times \mathbb{Z}/\ell \mathbb{Z}$, $v_\ell(p - 1) = 1$

We follow a similar strategy to Section 4.4

To calculate the probability for $\mathbb{Z}/\ell^e \mathbb{Z} \times \mathbb{Z}/\ell \mathbb{Z}$ we want to look at $GL_2(\mathcal{O}_{e+2})$ and this time we don't need to deal separately with small e because for every $e \geq 1$ we won't deal with elements with "top" reduced valuation.

We define the following quantities for every $k \geq 2$ and $D \in \mathcal{O}_k^\times$:

$$\begin{aligned}
N(k, D) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} \det(1 - M) = 0, \det(M) = D, \\ \gcd(1 - a, b, c, 1 - d) = \ell \end{array} \right. \right\} \\
M(k, D) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} v(\det(1 - M)) = k - 1, \det(M) = D, \\ \gcd(1 - a, b, c, 1 - d) = \ell \end{array} \right. \right\} \\
M(k) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} v(\det(1 - M)) = k - 1, \\ v(1 - \det(M)) = 1, \\ \gcd(1 - a, b, c, 1 - d) = \ell \end{array} \right. \right\} = \sum_{D \in \mathcal{O}_k^\times, v(1-D)=1} M(k, D)
\end{aligned}$$

Claim 14. For any $k \geq 2$ and $D \in \mathcal{O}_k^\times$ such that $v(1 - D) = 1$

$$N(k, D) = \ell^{2k-1}$$

Proof. Writing $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We are counting all matrices such that $ad - bc = D$, $1 - a - d + D = 0$ and $\min\{v(1-a), v(b), v(c), v(1-d)\} = 1$. For every a , d is uniquely determined by $d = 1 - a + D$, so we only need to count options for a, b, c . Furthermore, $bc = ad - D = ad - (a + d - 1) = (1 - a)(1 - d)$, so $v(b) \leq v((1 - a)(1 - d))$ and for every b there are $\ell^{v(b)}$ options for c (lemma 15 in [MS21]). Lastly, note that $1 - D = (1 - a) + (1 - d)$ so we have to following three cases contributing to $N(k, D)$:

1. $v(1 - a) = v(1 - d) = 1$
2. $v(1 - a) = 1, v(1 - d) \geq 2$
3. $v(1 - a) \geq 2, v(1 - d) = 1$

Now we count each case separately.

1. $v(1 - a) = v(1 - d) = 1$. Writing $a = a_0 + a_1\ell + \dots + a_{k-1}\ell^{k-1}$, the constraints $v(1 - a) = v(a - D) = v(1 - D) = 1$ translate into $a_0 = 1$ and $a_1 \neq 0, D_1$, thus a has $(\ell - 2)\ell^{k-2}$ options. From $bc = (1 - a)(1 - d)$ we get $v(bc) = 2$ and $v(b), v(c) \geq 1$, so if $k > 2$ then the only option is $v(b) = 1$, b has $\ell^{k-2}(\ell - 1)$ options and for each one c has ℓ options. If $k = 2$ then another option is that at least one of $b = 0$ and then c has ℓ more options, distinct from the previous options. Together the contribution to $N(k, D)$ is:

$$\underbrace{(\ell - 2)\ell^{k-2}}_{\text{options for } a} \underbrace{(\ell^{k-2}(\ell - 1) \cdot \ell + \delta_{2,k}\ell)}_{\text{options for } b, c} = \ell^{2k-3}(\ell - 1)(\ell - 2) + \delta_{2,k} \cdot \ell(\ell - 2)$$

2. By symmetric reasoning $a \leftrightarrow d$ this case contributes the same as case 3 below.
3. Denote $t := v(1 - a) \geq 2, s := v(b)$.

If $t = k$, then $a = 1$ and $1 - a - d + D = 0 \Rightarrow d = D$, so the number of options for a is 1.

If $t < k$, writing $a = a_0 + a_1\ell + \dots + a_{k-1}\ell^{k-1}$ the constraints translate to $a_0 = 1$, $a_1 = \dots = a_{t-1} = 0$, $a_t \neq 0$ and we get $v(1 - d) = v(a - D) = 1$ automatically, so the number of options for a is $(\ell - 1)\ell^{k-t-1}$.

From $bc = (1 - a)(1 - d)$ we get $v(b) \leq v(bc) = \min\{t + 1, k\}$. If $t + 1 < k$ then $s < t + 1$ because for $s = t + 1$ we get $v(c) = 0$. Furthermore, for each $s < t + 1$ b has $(\ell - 1)\ell^{k-s-1}$ options and for each one c has ℓ^s options so together the pair b, c has $(\ell - 1)\ell^{k-1}$ options.

If $t + 1 \geq k$ then $bc = 0$. For $s = k$ b has 1 option and c has ℓ^{k-1} options. For every $s < k$, b has $(\ell - 1)\ell^{k-s-1}$ options and for each one c has ℓ^s options so together the pair b, c has $(\ell - 1)\ell^{k-1}$ options

Together the contribution of this case is:

$$\begin{aligned}
& (\ell^{k-1} + \sum_{s=1}^{k-1} (\ell - 1)\ell^{k-1}) \left(\underbrace{1}_{t=k} + \underbrace{(\ell - 1) - \delta_{2,k}(\ell - 1)}_{t=k-1} \right) + \sum_{t=2}^{k-2} \sum_{s=1}^t (\ell - 1)\ell^{k-t-1}(\ell - 1)\ell^{k-1} = \\
& = \ell^k + (k - 1)(\ell - 1)\ell^k - \delta_{2,k}\ell^2(\ell - 1) + (\ell - 1)^2\ell^{2k-2} \sum_{t=2}^{k-2} t\ell^{-t} = \\
& = (k - 1)\ell^{k+1} - (k - 2)\ell^k - \delta_{2,k}\ell^2(\ell - 1) + (\ell - 1)^2\ell^{2k-2} \left(\frac{-(k - 1)\ell^4 + (k - 2)\ell^3 + \ell^k(2\ell - 1)}{\ell^{k+1}(\ell - 1)^2} + \delta_{2,k}\ell^{-1} \right) = \\
& = (k - 1)\ell^{k+1} - (k - 2)\ell^k + \ell^{k-3} \left(-(k - 1)\ell^4 + (k - 2)\ell^3 + \ell^k(2\ell - 1) \right) - \delta_{2,k}(\ell - 1)\ell = \\
& = \ell^{2k-3}(2\ell - 1) - \delta_{2,k}(\ell - 1)\ell
\end{aligned}$$

Now we sum all cases and finally get:

$$\begin{aligned}
N(k, D) &= \ell^{2k-3}(\ell - 1)(\ell - 2) + \delta_{2,k} \cdot \ell(\ell - 2) + 2 \cdot (2\ell^{2k-2} - \ell^{2k-3} - \delta_{2,k}(\ell - 1)\ell) = \\
&= \ell^{2k-2}(\ell + 1) - \delta_{2,k}\ell^2 = \begin{cases} \ell^3 & k = 2 \\ \ell^{2k-2}(\ell + 1) & k \geq 3 \end{cases}
\end{aligned}$$

□

Claim 15. For every $k \geq 3$ and $D \in \mathcal{O}_k^\times$ such that $v(D - 1) = 1$, denoting by $\pi : \mathcal{O}_k \rightarrow \mathcal{O}_{k-1}$ the projection, we have:

$$M(k, D) = \ell^3 N(k - 1, \pi(D)) - N(k, D)$$

Proof. Exactly the proof of Claim 8. We didn't assume anything on D there other than $D \in \mathcal{O}_k^\times$.

Note that for $k \geq 3$ then two matrices $M_1, M_2 \in GL_2(\mathcal{O}_k)$ in the same fiber of the projection $GL_2(\mathcal{O}_k) \rightarrow GL_2(\mathcal{O}_{k-1})$ differ by a matrix multiple of ℓ^{k-1} and so a coefficient of $1 - M_1$ is divisible by ℓ if and only if the corresponding coefficient of $1 - M_2$ is divisible by ℓ . $\square$

Corollary 7. *For every $k \geq 3$ and $D \in \mathcal{O}_k^\times$ such that $v(D - 1) = 1$*

$$\begin{aligned} M(k, D) &= \ell^3(\ell^{2k-4}(\ell + 1) - \delta_{2,k-1}\ell^2) - (\ell^{2k-2}(\ell + 1) - \delta_{2,k}\ell^2) = \ell^{2k-1}(\ell + 1) - \ell^{2k-2}(\ell + 1) - \delta_{3,k}\ell^5 = \\ &= \ell^{2k-2}(\ell + 1)(\ell - 1) - \delta_{3,k}\ell^5 = \begin{cases} \ell^4(\ell^2 - \ell - 1) & k = 3 \\ \ell^{2k-2}(\ell + 1)(\ell - 1) & k \geq 4 \end{cases} \end{aligned}$$

Corollary 8. *For every $k \geq 3$*

$$\begin{aligned} M(k) &= \sum_{D \in \mathcal{O}_k^\times, v(1-D)=1} M(k, D) = \underbrace{\#\{D \in \mathcal{O}_k^\times, v(1-D)=1\}}_{=(\ell-1)\ell^{k-2}} \cdot (\ell^{2k-2}(\ell + 1)(\ell - 1) - \delta_{3,k}\ell^5) = \\ &= \ell^{3k-4}(\ell + 1)(\ell - 1)^2 - \delta_{3,k}\ell^6(\ell - 1) \end{aligned}$$

Claim 16. *For every $e \geq 1$:*

$$\mu \left(\left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathbb{Z}_\ell) \mid \begin{array}{l} v_\ell(\det(M) - 1) = 1, \\ v_\ell(\det(1 - M)) = e + 1, \\ \gcd(1 - a, b, c, 1 - d) = \ell \end{array} \right\} \right) = \begin{cases} \frac{\ell^2 - \ell - 1}{\ell^4(\ell^2 - 1)} & e = 1 \\ \frac{1}{\ell^{e+3}} & e \geq 2 \end{cases}$$

Proof. As in the proof of Claim 10. Take $k = e + 2$ and denote this set by X . So X is "detectable modulo ℓ^k " and $M(k)$ exactly counts the cosets modulo ℓ^k we want, so:

$$\begin{aligned} \mu(X) &= \frac{M(e+2)}{\#GL_2(\mathcal{O}_{e+2})} = \frac{\ell^{3e+2}(\ell + 1)(\ell - 1)^2 - \delta_{1,e}\ell^6(\ell - 1)}{\ell^{4(e+1)}(\ell^2 - 1)(\ell^2 - \ell)} = \\ &= \frac{\ell^{3e-4}(\ell + 1)(\ell - 1) - \delta_{1,e}}{\ell^{4e-1}(\ell^2 - 1)} = \frac{1}{\ell^{e+3}} - \frac{\delta_{1,e}}{\ell^3(\ell^2 - 1)} = \begin{cases} \frac{\ell^2 - \ell - 1}{\ell^3(\ell^2 - 1)} & e = 1 \\ \frac{1}{\ell^{e+3}} & e \geq 2 \end{cases} \end{aligned}$$

$\square$

Corollary 9. *The above claim concludes neatly the result that for all $e \geq 0$*

$$\mathbb{P} \left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e\mathbb{Z} \times \mathbb{Z}/\ell\mathbb{Z} \mid v_\ell(p - 1) = 1 \right) = \begin{cases} \frac{\ell^2 - \ell - 1}{\ell^3(\ell^2 - 1)} & e = 1 \\ \frac{1}{\ell^{e+2}} & e \geq 2 \end{cases}$$

given that ρ_ℓ is surjective.

Proof. Dirichlet's theorem on arithmetic progressions implies that the density of primes satisfying $v_\ell(p-1) = 1$ is $\frac{1}{\ell}$. Dividing the joint probability we calculated before by this will give us the conditional probabilities we want. $\square$

4.7 Cyclic case, $G = \mathbb{Z}/\ell^e\mathbb{Z}$, $v_\ell(p-1) = 2$

For every $k \geq 2$, $D \in \mathcal{O}_k^\times$: We define the following quantities:

$$\begin{aligned} N(k, D) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} \det(1-M) = 0, \det(M) = D, \\ \gcd(1-a, b, c, 1-d) = 1 \end{array} \right. \right\} \\ M(k, D) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} v(\det(1-M)) = k-1, \det(M) = D, \\ \gcd(1-a, b, c, 1-d) = 1 \end{array} \right. \right\} \\ M(k) &:= \# \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathcal{O}_k) \left| \begin{array}{l} v(\det(1-M)) = k-1, \\ v(1-\det(M)) = 2, \\ \gcd(1-a, b, c, 1-d) = 1 \end{array} \right. \right\} = \sum_{D \in \mathcal{O}_k^\times, v(1-D)=2} M(k, D) \end{aligned}$$

Claim 17. For any $k \geq 3$ and $D \in \mathcal{O}_k$ such that $v(1-D) = 2$

$$N(k, D) = \ell^{2k-2}(\ell-1)(\ell+1)$$

Proof. Writing $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. We are counting all matrices such that $ad - bc = D$, $1 - a - d + D = 0$ and $\min\{v(1-a), v(b), v(c), v(1-d)\} = 0$. For every a , d is uniquely determined by $d = 1 - a + D$, so we only need to count options for a, b, c . Furthermore, $bc = ad - D = ad - (a + d - 1) = (1-a)(1-d)$, so $v(b) \leq v((1-a)(1-d))$ and for every b there are $\ell^{v(b)}$ options for c (lemma 15 in [MS21]). Lastly, note that $1 - D = (1-a) + (1-d)$ so we have to following four cases contributing to $N(k, D)$:

1. $v(1-a) = v(1-d) = 0$
2. $v(1-a) = v(1-d) = 1$
3. $v(1-a) = v(1-d) = 2$
4. $v(1-a) = 2, v(1-d) \geq 3$
5. $v(1-a) \geq 3, v(1-d) = 2$

Now we count each case separately.

1. $v(1-a) = 0$ so the gcd is 1 always. Furthermore $v(b) = 0$ and there is 1 option for c for any b . Because $1-d = a - D$ and $v(1-D) = 2$ then $\overline{D} = 1 \in \mathbb{Z}_\ell/\ell\mathbb{Z}_\ell$ and $\overline{a} \neq 1 \in \mathbb{Z}_\ell/\ell\mathbb{Z}_\ell$. So we have $\ell-1$ options for the first coordinate of a (the projection to $\mathbb{Z}_\ell/\ell\mathbb{Z}_\ell$) and ℓ options for

each of the other coordinates and in general the contribution of this case to $N(k, D)$ is:

$$\underbrace{(\ell-1)\ell^{k-1}}_{\text{options for } a} \underbrace{(\ell-1)\ell^{k-1}}_{\text{options for } b} = \ell^{2k-2}(\ell-1)^2$$

2. $v(1-a) = v(1-d) = 1$, so from $bc = (1-a)(1-d)$ we get $v(bc) = 2$ and we need at least one of b, c to have valuation 0. So the options are either $v(b) = 0$ and $v(c) = 2$ or the opposite, from symmetric reasoning it is enough to multiply by 2 the count of the first option. Writing $a = a_0 + a_1\ell + \dots + a_{k-1}\ell^{k-1}$, the constraints $v(1-a) = v(a-D) = 1$, $v(1-D) = 2$ translate into $a_0 = 1$ and $a_1 \neq 0$, thus a has $(\ell-1)\ell^{k-2}$ options and assuming $v(b) = 0$, b has $(\ell-1)\ell^{k-1}$ options so together the contribution to $N(k, D)$ is:

$$2 \underbrace{(\ell-1)\ell^{k-2}}_{\text{options for } a} \underbrace{(\ell-1)\ell^{k-1}}_{\text{options for } b} = 2\ell^{2k-3}(\ell-1)^2$$

3. $v(1-a) = v(1-d) = 2$, so from $bc = (1-a)(1-d)$ we get $v(bc) = \min\{k, 4\}$ and we need at least one of b, c to have valuation 0. So the options are either $v(b) = 0$ and $v(c) = \min\{k, 4\}$ or the opposite, from symmetric reasoning it is enough to multiply by 2 the count of the first option. Writing $a = a_0 + a_1\ell + \dots + a_{k-1}\ell^{k-1}$, the constraints $v(1-a) = v(a-D) = v(1-D) = 2$ translate into $a_0 = 1$, $a_1 = 0$ and $a_2 \neq 0, D_2$, thus a has $(\ell-2)\ell^{k-3}$ options. Assuming $v(b) = 0$, b has $(\ell-1)\ell^{k-1}$ options and for each one c has 1 option so together the contribution to $N(k, D)$ is:

$$2 \underbrace{(\ell-2)\ell^{k-3}}_{\text{options for } a} \underbrace{(\ell-1)\ell^{k-1}}_{\text{options for } b} = 2\ell^{2k-4}(\ell-1)(\ell-2)$$

4. By symmetric reasoning $a \leftrightarrow d$ this case contributes the same as case 4 below.
5. Denote $t := v(1-a) \geq 3$. First of all, $\min\{v(b), v(c)\} = 0$ and $bc = ad - D$ which isn't a unit so exactly one of b, c is a unit, assume for now that $v(b) = 0$ and we'll double the result. So there are $(\ell-1)\ell^{k-1}$ options for b and for each one there is a single option for c .
If $t = k$ then $a = 1$, from $1 - a - d + D = 0$ we get $d = D$, so given b, c there is one option for a and d .
If $t < k$, writing $a = a_0 + a_1\ell + \dots + a_{k-1}\ell^{k-1}$ the constraints translate to $a_0 = 1$, $a_1 = \dots = a_{t-1} = 0$, $a_t \neq 0$ and we get $v(1-d) = v(a-D) = 2$ automatically, so the number of options

for a is $(\ell - 1)\ell^{k-t-1}$. Together the contribution of this case is:

$$\begin{aligned} & \underbrace{2(\ell - 1)\ell^{k-1}}_{\text{options for } b} \cdot \underbrace{\left(1 + \sum_{t=3}^{k-1} (\ell - 1)\ell^{k-t-1}\right)}_{\text{options for } a} = 2(\ell - 1)\ell^{k-1} \left(1 + (\ell - 1) \sum_{j=0}^{k-4} \ell^j\right) = \\ & = 2(\ell - 1)\ell^{k-1} (1 + \ell^{k-3} - 1) = 2\ell^{2k-4}(\ell - 1) \end{aligned}$$

Now we sum all cases and finally get:

$$\begin{aligned} N(k, D) &= \ell^{2k-2}(\ell - 1)^2 + 2\ell^{2k-3}(\ell - 1)^2 + 2\ell^{2k-4}(\ell - 1)(\ell - 2) + 2 \cdot 2\ell^{2k-4}(\ell - 1) = \\ &= (\ell - 1) (\ell^{2k-1} - \ell^{2k-2} + 2\ell^{2k-2} - 2\ell^{2k-3} + 2\ell^{2k-3} - 4\ell^{2k-4} + 4\ell^{2k-4}) = \\ &= \ell^{2k-2}(\ell - 1)(\ell + 1) \end{aligned}$$

□

Claim 18. $\forall k \geq 4$, denoting by $\pi : \mathcal{O}_k \rightarrow \mathcal{O}_{k-1}$ the projection, we have:

$$M(k, D) = \ell^3 N(k - 1, \pi(D)) - N(k, D)$$

Proof. Exactly the proof of Claim 8. We didn't assume anything on D there other than $D \in \mathcal{O}_k^\times$. Note that for $k \geq 3$ then two matrices $M_1, M_2 \in GL_2(\mathcal{O}_k)$ in the same fiber of the projection $GL_2(\mathcal{O}_k) \rightarrow GL_2(\mathcal{O}_{k-1})$ differ by a matrix multiple of ℓ^{k-1} and so a coefficient of $1 - M_1$ is divisible by ℓ if and only if the corresponding coefficient of $1 - M_2$ is divisible by ℓ . □

Corollary 10. $\forall k \geq 4$:

$$M(k, D) = \ell^3 N(k - 1, \pi(D)) - N(k, D) = \ell^{2k-3}(\ell - 1)(\ell + 1) - \ell^{2k-2}(\ell - 1)(\ell + 1) = \ell^{2k-2}(\ell - 1)^2(\ell + 1)$$

Corollary 11. $\forall k \geq 4$:

$$\begin{aligned} M(k) &= \sum_{D \in \mathcal{O}_k^\times, v(1-D)=2} M(k, D) = \\ &= \underbrace{\#\{D \in \mathcal{O}_k^\times, v(1-D)=2\}}_{=\ell^{k-2}(\ell-1)} \cdot \ell^{2k-2}(\ell - 1)^2(\ell + 1) = \ell^{3k-4}(\ell - 1)^3(\ell + 1) \end{aligned}$$

Claim 19. *For every $e \geq 3$,*

$$\mu \left(\left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathbb{Z}_\ell) \left| \begin{array}{l} v_\ell(\det(M) - 1) = 2, \\ v_\ell(\det(1 - M)) = e, \\ \gcd(1 - a, b, c, 1 - d) = 1 \end{array} \right. \right\} \right) = \frac{\ell - 1}{\ell^{e+2}}$$

Proof. Denote this set by X . So X is "detectable modulo ℓ^k " for $k = e + 1$ and $M(e + 1)$ exactly counts the cosets we want, so

$$\mu(X) = \frac{M(e + 1)}{\#GL_2(\mathcal{O}_{e+1})} = \frac{\ell^{3e-1}(\ell - 1)^3(\ell + 1)}{\ell^{4e}(\ell^2 - 1)(\ell^2 - \ell)} = \frac{\ell - 1}{\ell^{e+2}}$$

□

Corollary 12. *The above claim concludes neatly the result that for all $e \geq 3$*

$$\mathbb{P} \left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z} / \ell^e \mathbb{Z} \mid v_\ell(p - 1) = 2 \right) = \frac{\ell - 1}{\ell^e}$$

given that ρ_ℓ is surjective.

Proof. Dirichlet's theorem on arithmetic progressions implies that the density of primes satisfying $v_\ell(p - 1) = 2$ is $\frac{1}{\ell^2}$. Dividing the joint probabilities we calculated before by this will give us the conditional probabilities we want. □

5 Further Discussion

5.1 Cohen-Lenstra Heuristics

In Their 1984 paper [CL84], Cohen and Lenstra gave heuristics to explain experimental observations they had regarding the class groups of quadratic fields. They explained heuristically why its reasonable to expect that some finite abelian group G will appear as a class group with probability proportional to $\frac{1}{\#\text{Aut}(G)}$. Modern development generalized these heuristics under the observation that the objects with the Cohen-Lenstra type distribution might be in a different category, for example finite abelian groups with a (certain type of) pairing.

5.2 Summary of computational results and conjecture

In our attempt to reinterpret our results in the context of Cohen-Lenstra heuristics, we found use of the following invariant

Definition 9. Given an abelian group G , we define its exterior power $\wedge^2 G$ to be $G^{\otimes 2}/I$ where $I \leq G^{\otimes 2}$ is the subgroup consisting of elements of the form $g \otimes g$. This is the general construction of exterior power in the category of modules, when looking at G as a $\mathbb{Z}$ -module.

It is known that

Claim 20. *For every $1 \leq e_1 \leq e_2$:*

$$\begin{aligned} \left| \wedge^2 \left(\mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z} \right) \right| &= \ell^{e_1} \\ \left| \wedge^2 \mathbb{Z}/\ell^{e_1}\mathbb{Z} \right| &= 1 \end{aligned}$$

and for every $1 \leq e_1 < e_2$:

$$\begin{aligned} \left| \text{Aut} \left(\mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z} \right) \right| &= \ell^{3e_1+e_2-2}(\ell-1)^2 \\ \left| \text{Aut} \left(\mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_1}\mathbb{Z} \right) \right| &= \ell^{4e_1-3}(\ell-1)(\ell^2-1) \\ \left| \text{Aut} \left(\mathbb{Z}/\ell^{e_1}\mathbb{Z} \right) \right| &= \varphi(\ell^{e_1}) \end{aligned}$$

Proof. For proof see general treatment in [Woo14]: The first two equations are specific cases of a general formula in subsection 2.4 and the last three equations are specific cases of lemma 7.2. $\square$

The following is a list of the computational results from Section 4 rewritten as formulas of invariants of the fixed group in each case (which we denote by G on the right hand side):

$$\mathbb{P}(E(\mathbb{F}_p)[\ell^\infty] = 0) = 1 - \frac{\ell^2}{(\ell-1)^2(\ell+1)} \quad (1)$$

$$\mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e \mathbb{Z} \mid v_\ell(p-1) = 0\right) = \begin{cases} \frac{\ell-2}{\ell-1} = \frac{(\ell-2)|\wedge^2 G|}{(\ell-1)|\text{Aut}(G)|} & e = 0 \\ \frac{1}{\ell^e} = \frac{(\ell-1)|\wedge^2 G|}{\ell|\text{Aut}(G)|} & e \geq 1 \end{cases} \quad (2)$$

$$\mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e \mathbb{Z} \mid v_\ell(p-1) = 1\right) = \begin{cases} 1 - \frac{\ell}{\ell^2-1} = \frac{(\ell^2-\ell-1)|\wedge^2 G|}{(\ell^2-1)|\text{Aut}(G)|} & e = 0 \\ \frac{\ell-1}{\ell^{e+1}} = \frac{(\ell-1)^2|\wedge^2 G|}{\ell^2|\text{Aut}(G)|} & e \geq 1 \end{cases} \quad (3)$$

$$\mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e \mathbb{Z} \times \mathbb{Z}/\ell \mathbb{Z} \mid v_\ell(p-1) = 1\right) = \begin{cases} \frac{\ell^2-\ell-1}{\ell^3(\ell^2-1)} = \frac{(\ell^2-\ell-1)(\ell-1)|\wedge^2 G|}{\ell^3|\text{Aut}(G)|} & e = 1 \\ \frac{1}{\ell^{e+2}} = \frac{(\ell-1)^2|\wedge^2 G|}{\ell^2|\text{Aut}(G)|} & e \geq 2 \end{cases} \quad (4)$$

$$\mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^e \mathbb{Z} \mid v_\ell(p-1) = 2\right) = \begin{cases} \frac{\ell-1}{\ell^e} = \frac{(\ell-1)^2|\wedge^2 G|}{\ell|\text{Aut}(G)|} & e \geq 3 \end{cases} \quad (5)$$

This can be seen as evidence that the conditional probability when fixing $v_\ell(p-1)$ follows a Cohen-Lenstra type distribution where the probability of a ("legal") group G is proportional to $\frac{|\wedge^2 G|}{|\text{Aut}(G)|}$. We conjecture that this happens in general:

Conjecture 1. *Given an elliptic curve E over $\mathbb{Q}$ without CM and a prime integer ℓ such that the ℓ -adic representation $\rho_\ell : G_{\mathbb{Q}} \rightarrow GL_2(\mathbb{Z}_\ell)$ is surjective, there exists a constant $C_{\ell,t} > 0$ depending only on ℓ and t such that for large enough $e_1 \leq e_2$:*

$$\mathbb{P}\left(E(\mathbb{F}_p)[\ell^\infty] \cong \mathbb{Z}/\ell^{e_1} \mathbb{Z} \times \mathbb{Z}/\ell^{e_2} \mathbb{Z} \mid v_\ell(p-1) = t\right) = C_{\ell,t} \frac{|\wedge^2 (\mathbb{Z}/\ell^{e_1} \mathbb{Z} \times \mathbb{Z}/\ell^{e_2} \mathbb{Z})|}{|\text{Aut}(\mathbb{Z}/\ell^{e_1} \mathbb{Z} \times \mathbb{Z}/\ell^{e_2} \mathbb{Z})|}$$

Furthermore, there seems to be a discrepancy for groups with small order, whose explanation remains unknown. This on top of the appearance of $|\wedge^2 G|$ in the formula hint that we are missing some extra structure not taken into account.

5.3 Testing results

After computing the probabilities above, we run a simulation with Sage. We take the elliptic curve $E : y^2 = x^3 + x + 1$. It does not have CM and its mod 5 representation is surjective and so by a theorem of Serre also its 5-adic representation is surjective (note that E has good reduction mod 5). We went through the first 250,000 primes and approximated the values we calculated in Section 4. We compile the final results in the following table, where in each row the left column denotes values for e_1 , e_2 and t , the right column is the value from our computation for the probability of $E(\mathbb{F}_p)[5^\infty]$ to be $\mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z}$ conditional on $v_5(p-1) = t$ and the middle column is the result from our simulation. Appendix A contains the code used for this simulation.

(0, 0, 0)	140831/187515 $\approx$ 0.75104	3/4 $\approx$ 0.75000
(0, 0, 1)	12449/62505 $\approx$ 0.19917	1/5 $\approx$ 0.20000
(0, 0, 2)	167/4167 $\approx$ 0.04008	1/25 $\approx$ 0.04000
(0, 0, 3)	290/37503 $\approx$ 0.00773	1/125 $\approx$ 0.00800
(0, 0, 4)	298/187515 $\approx$ 0.00159	1/625 $\approx$ 0.00160
(0, 0, 5)	11/37503 $\approx$ 0.00029	1/3125 $\approx$ 0.00032
(0, 0, 6)	4/62505 $\approx$ 0.00006	1/15625 $\approx$ 0.00006
(1, 0, 0)	39663/49988 $\approx$ 0.79345	19/24 $\approx$ 0.79167
(1, 0, 1)	3977/24994 $\approx$ 0.15912	4/25 $\approx$ 0.16000
(1, 0, 2)	1621/49988 $\approx$ 0.03243	4/125 $\approx$ 0.03200
(1, 0, 3)	297/49988 $\approx$ 0.00594	4/625 $\approx$ 0.00640
(1, 0, 4)	51/49988 $\approx$ 0.00102	4/3125 $\approx$ 0.00128
(1, 0, 5)	13/49988 $\approx$ 0.00026	4/15625 $\approx$ 0.00026
(1, 1, 1)	149/24994 $\approx$ 0.00596	19/3000 $\approx$ 0.00633
(1, 1, 2)	73/49988 $\approx$ 0.00146	1/625 $\approx$ 0.00160
(1, 1, 3)	13/49988 $\approx$ 0.00026	1/3125 $\approx$ 0.00032
(2, 0, 0)	7941/10006 $\approx$ 0.79362	N/A
(2, 0, 1)	810/5003 $\approx$ 0.16190	N/A
(2, 0, 2)	144/5003 $\approx$ 0.02878	N/A
(2, 0, 3)	29/5003 $\approx$ 0.00580	4/125 $\approx$ 0.03200
(2, 0, 4)	13/10006 $\approx$ 0.00130	4/625 $\approx$ 0.00640
(2, 1, 1)	63/10006 $\approx$ 0.00630	N/A
(2, 1, 2)	15/10006 $\approx$ 0.00150	N/A

5.4 The Moment Problem

Although clever computations might make the ideas of Section 4 more general, it might be easier to come by general results by calculating moments and recovering the distribution.

It is a classical problem in probability theory to determine whether for a given sequence $(M_n)_{n=1}^\infty$ there exists a distribution whose n -th moment is M_n for every n , i.e a probability measure μ such that $M_n = \int x^n d\mu(x)$. Furthermore, if such a probability exists, is it unique? could it be expressed or constructed by some algorithm using $(M_n)_{n=1}^\infty$?

This idea was adapted and used ad-hoc, for example in [Woo14] and [LST20], and culminating in a general formalization as a framework for density problems in arithmetic statistics in [SW22].

The ideas work in every nice enough category C , but for simplicity we'll take C to be the category of finite groups.

Definition 10 (G -Moment of a probability measure ν). Given a finite group G and a probability measure ν on $C_{/\simeq}$ (isomorphism classes of C), we define the G -moment of ν to be $\int_{X \in C_{/\simeq}} \# \text{Sur}(X, G) d\nu$

In this context, the moment problem asks given a set $\{M_G\}_{G \in C_{/\simeq}}$, does there exists a probability measure on $C_{/\simeq}$ whose G -moment is M_G for every G ? If so, is it unique and could it be reconstructed from $\{M_G\}_{G \in C_{/\simeq}}$? Vaguely speaking, the answer is that if $\{M_G\}_{G \in C_{/\simeq}}$ satisfy some analytic conditions (such as, "they don't grow too fast") then the answers to the above questions are affirmative.

A Simulation Code

This is the sage code written for the simulation described in Section 5.3, where the table is the output of `Simulate(1,1,5,[0,1,2],250000,10)`.

```

1 def Simulate(a, b, ell, tee_list, num_of_primes, minimal_cutoff):
2     E = EllipticCurve([a,b]) # E:  $y^2 = x^3 + ax + b$ 
3     primesCounter = {tee:0 for tee in tee_list}
4     groupsCounter = {}
5     assert not E.has_cm(), "E has CM!"
6     assert E.galois_representation().is_surjective(ell), f"The galois representation is not
       surjective at ell={ell}"
7
8     for p in primes_first_n(num_of_primes):
9         tee = (p-1).valuation(ell)
10        # If  $v_\ell(p-1) = t$  and  $p$  is a prime of good reduction
11        if tee in tee_list and not p.divides(E.conductor()):
12            primesCounter[tee] += 1
13
14        # This extracts the  $(x,y)$  such that  $E_p(F_p) \cong \mathbb{Z}/x\mathbb{Z} \times \mathbb{Z}/y\mathbb{Z}$ 
15        generator_orders = E.reduction(p).abelian_group().generator_orders()
16        generator_orders = (1,)*(2-len(generator_orders)) + generator_orders
17
18        # This converts those  $(x,y)$  to the exponents  $(e_1,e_2)$  such
19        # that  $0 \leq e_1 \leq e_2$  and  $E_p(F_p)[\ell^\infty] \cong \mathbb{Z}/\ell^{e_1}\mathbb{Z} \times \mathbb{Z}/\ell^{e_2}\mathbb{Z}$ 
20        exponents = tuple(sorted([x.valuation(ell) for x in generator_orders]))
21        groupsCounter[(tee,)+exponents] = groupsCounter.get((tee,)+exponents,0) + 1
22
23    for key in sorted(groupsCounter):
24        if groupsCounter[key] >= minimal_cutoff:
25            ratio = groupsCounter[key]/primesCounter[key[0]]
26            print(f"${key}$ & ${ratio}$ \\approx {float(ratio):.5f}$ & ${TheoreticalVal(*key,ell)}$
              \\")
27
28 def TheoreticalVal(tee,e_1,e_2,ell):
29     value = 0
30     if tee == 0:
31         value = (ell-2)/(ell-1) if e_2 == 0 else 1/(ell^e_2)
32     elif tee == 1:
33         if e_1 == 0:
34             value = 1-ell/(ell^2-1) if e_2 == 0 else (ell-1)/(ell^(e_2+1))
35         else:
36             value = (ell^2-ell-1)/(ell^3*(ell^2-1)) if e_2 == 1 else 1/(ell^(e_2+2))
37     elif tee == 2 and e_1 == 0 and e_2 >= 3:
38         value = (ell-1)/(ell^e_2)
39     return("N/A" if value == 0 else f"{value}$ \\approx {float(value):.5f}")

```

References

- [CL84] H. Cohen and H. W. Lenstra. “Heuristics on class groups of number fields”. In: *Number Theory Noordwijkerhout 1983*. Ed. by Hendrik Jager. Berlin, Heidelberg: Springer Berlin Heidelberg, 1984, pp. 33–62. ISBN: 978-3-540-38906-4.
- [LST20] Michael Lipnowski, Will Sawin, and Jacob Tsimerman. *Cohen-Lenstra heuristics and bilinear pairings in the presence of roots of unity*. 2020. DOI: 10.48550/ARXIV.2007.12533. URL: <https://arxiv.org/abs/2007.12533>.
- [MS21] Nathanaël Munier and Ari Shnidman. *Sandpile groups of supersingular isogeny graphs*. 2021. DOI: 10.48550/ARXIV.2111.10389. URL: <https://arxiv.org/abs/2111.10389>.
- [Ser68] Jean-Pierre Serre. *Abelian l -adic representations and elliptic curves : McGill University lecture notes*. eng. New York: W. A. Benjamin, 1968.
- [Sil09] Joseph H. Silverman. *The arithmetic of elliptic curves*. eng. 2nd ed. Graduate texts in mathematics ; 106. New York: Springer, 2009. ISBN: 9780387094939.
- [SW22] Will Sawin and Melanie Matchett Wood. *The moment problem for random objects in a category*. 2022. DOI: 10.48550/ARXIV.2210.06279. URL: <https://arxiv.org/abs/2210.06279>.
- [Woo14] Melanie Matchett Wood. *The distribution of sandpile groups of random graphs*. 2014. DOI: 10.48550/ARXIV.1402.5149. URL: <https://arxiv.org/abs/1402.5149>.